| METRIC |

MIL-M-24791
25 May 1995
SUPERSEDING
MIL-T-24721(NAVY)
27 June 1989
MIL-R-24720(NAVY)
7 July 1989

MILITARY SPECIFICATION

MODULE, FIBER OPTIC, TRANSMITTER OR RECEIVER, DIGITAL
GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification describes digital fiber optic transmitters and receivers. A digital transmitter is an optoelectronic transducer which converts a binary logic electrical input signal to a modulated optical output signal. A digital receiver is an optoelectronic transducer which converts a modulated optical input signal to a binary logic electrical output signal. A transmitter-receiver pair may be used to transfer information between two points through an optical fiber.

1.2 Reference to "item to be procured". This specification and the associated detail specification define the technical requirements for an item to be acquired, either a digital fiber optic transmitter or receiver, as briefly described in 1.1 herein, the item is referred to generically as "module" or "device".

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-H-38534	-	Hybrid Microcircuits, General Specification for
MIL-F-49291/6	-	Fiber, Optical, 62.5/125 Micrometers, Radiation Hardened (Metric)
MIL-M-55565	-	Microcircuits, Packaging

STANDARDS

FEDERAL

FED-STD-209	-	Clean Room And Work Station Requirements, Controlled Environment
FED-STD-1037	-	Glossary Of Telecommunication Terms

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Sea Systems Command, SEA 05Q42, Washington, DC 20362-5160, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6030

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-883	-	Test Methods And Procedures For Microelectronics
MIL-STD-1285	-	Marking Of Electrical And Electronic Parts

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation, and the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION

EIA-440	-	Fiber Optic Terminology
EIA-455-6	-	Cable Retention Test Procedure For Fiber Optic Cable Interconnecting Devices
EIA-455-36	-	Twist Test For Fiber Optic Connecting Devices
JEDEC108	-	Distributor Requirements For Handling Electrostatic-Discharge Sensitive (ESDS) Circuits
JESD19	-	General Standard For Statistical Process Control (SPC)

(Application for copies should be addressed to the Electronic Industry Association, Engineering Department, 2001 Eye Street NW, Washington DC 20006.)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

ANSI/IEEE STD 100-1988	-	Standard Dictionary Of Electrical And Electronics Terms
IEEE STD 194-1977	-	Pulse Terms And Definitions
ANSI/IEEE STD 260-1978	-	IEEE Standard Letter Symbols For Units Of Measurement
ANSI/IEEE STD 280-1985	-	IEEE Standard Letter Symbols For Quantities Used In Electrical Science And Electrical Engineering

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, 445 Hoes Lane, Piscataway, NJ 08854-4150.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z136.2-1988	-	Laser Safety In Fiber Optics

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3308.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specifications or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General.

3.1.1 Associated detail specification. Individual requirements shall be as stated herein and in the associated detail specification. In the event of conflict between this document and the associated detail specification, the associated detail specification shall take precedence.

3.1.2 Concurrent application of requirements. Unless stated otherwise, all requirements of this specification and the associated detail specification apply concurrently.

3.1.3 Hierarchy. Reference to a paragraph implies reference to all its subparagraphs.

3.2 Functional.

3.2.1 Device functions. A transmitter device or receiver device shall provide the respective function specified in its subparagraph to this paragraph and shall satisfy the functional requirements of its associated detail specification.

3.2.1.1 Transmitter. A transmitter shall supply a modulated optical carrier (see 3.2.2.1) to its optical interface (see 3.2.2.2). The top and base pattern of the output envelope waveform shall be controlled by the signal pattern at the DATA input(s).

3.2.1.2 Receiver. A receiver shall recover binary data from an optical signal (see 3.2.2) supplied to its optical interface (see 3.2.2.2). The pattern at the DATA output(s) shall be controlled by the top and base pattern of the input envelope waveform.

3.2.2 Optical signal. The term "optical signal" refers to the modulated optical carrier. Optical modulation, interface, and power requirements shall be as specified in the subparagraphs to this paragraph.

3.2.2.1 Carrier modulation. Optical carrier modulation shall be binary intensity modulation (see 6.3.2.1).

3.2.2.2 Interface. Optical signal requirements of the associated detail specification sheets apply at the optical interface (see 6.3.2.10).

3.2.2.2.1 Reference fiber. The reference fiber (see 6.3.2.10) is specified in the associated detail specification.

3.2.2.3 Power. Unless otherwise specified in the associated detail specification, values of optical power at the optical interface (see 3.2.2.2) are averages over intervals for which the envelope waveform pattern has a duty factor of 0.50 ±0.020.

3.2.3. Short circuit protection. Electrical terminals (see 3.4.2.5) may be designated "short circuit Protected", as specified in the associated detail specification. Device performance shall not become non-compliant with the requirements herein and the associated detail specification after a short circuit connection is made and subsequently removed from those designated terminal(s).

3.3 Environmental. Functional requirements shall apply for device environments specified by the parameter values contained in table I.

3.4 Design and construction. The materials and constructions employed to fabricate the required device and the form and placement of its component parts shall be as specified herein.

3.4.1 Materials.

3.4.1.1 Hazardous. Materials producing toxic, corrosive, explosive or other hazards with the device in its intended applications shall not be used.

3.4.1.2 External. Cases (see 3.4.2.4) shall be fabricated from glass, metal or ceramic materials only. Combinations of these materials may be used. External parts, elements or coatings including markings shall not blister, crack, outgas, soften, or flow so as to compromise required performance and shall be inherently non-nutrient to fungus.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

TABLE I. Environmental requirements.

Parameters			Min	Max	Units	Conditions
Temperature	Case operating:	Range 1	-55	125	°C	see 6.3.2.3; 1/
		Range 2	-28	85	°C	1/
		Range 3	0	85	°C	1/
	Ambient storage		-28	85	°C	2/
Vibration	10 Hz to 2 KHz		0	10	g	Any direction
Relative humidity			5%	95%		Non-condensing
Impact shock	0.5 Ms duration		0	1500	g	Any direction

1/ Applicable range shall be as designated in the associated detail specification sheets; the difference between case and ambient (still air) shall be no greater than 25 °C.

2/ Applies to range 2 and 3 only and for time up to five years; functional requirements apply when case temperature reenters operating range.

3.4.1.3 Internal. The use of organic or polymeric materials (lacquers, varnishes, coatings, adhesives, greases, etc.) or getter materials inside the case shall be subject to procuring activity approval. Polymer impregnations (backfill, docking, coating, or other uses of polymeric materials to effect, improve, or repair the seal) of the case shall not be used. No adhesive or polymeric materials shall be used for case lid attachment, sealing or rework. Only welding, soldering, or glass sealing are permitted for those purposes. Packages with the lead frame attached to the body using a glass frit shall have glass on the mating surfaces only. The inside surfaces shall not be coated with the glass seal.

3.4.1.3.1 Glassivation. Unless otherwise specified, integrated circuits used in the device package shall be glassivated. Glassivation material and thickness data shall be available for procuring activity review and are subject to procuring activity approval.

3.4.2 Package. The package includes the case, optical connection and electrical terminals.

3.4.2.1 Style. The package style identifies the type of case and optical connections to be used. Styles are described in Appendix C. Styles shall be in accordance with the associated detail specification.

3.4.2.2 Seal. Packages shall be hermetically sealed. The leak rate shall not exceed 5×10^{-7} atm-cm^3/s.

3.4.2.3 Water content. Internal water vapor content shall not exceed 5,000 ppm at + 100 °C.

3.4.2.4 Case (see 6.3.2.2).

3.4.2.4.1 Type. The type shall be as specified in Appendix C for the required package style (see 3.4.2.1).

3.4.2.4.2 Finish. External metal surfaces shall be corrosion resistant. The package body finish requirements of MIL-H-38534 shall apply.

3.4.2.5 Electrical terminals.

3.4.2.5.1 Assignment. Assignment refers to the association of specific terminals with internal device functions (such as input, output, power supply voltage, ground, etc,...). Assignment shall be as specified in the associated detail specification.

3.4.2.5.2 Type. The terminal type shall be in accordance with the associated detail specification.

3.4.2.5.3 Materials. Materials shall be in accordance with the lead or terminal material requirements of MIL-H-38534.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

3.4.2.5.4 Strength. Handling associated with inspections and shipping and insertion and removal from sockets intended to receive such devices, performed in accordance with accepted procedures for such operations, shall not change electrical terminal positions to a degree requiring correction prior to insertion in a printed circuit board or socket intended to receive the device.

3.4.2.5.5 Finish. Electrical terminal finishes shall be in accordance with the lead finish requirements of MIL-H-38534.

3.4.2.5.6 Solderability. Terminal design shall be capable of supporting a lead/tin solder joint maximizing bond strength and electrical conductivity without requiring the use of corrosive fluxes and minimizing heat transfer to internal components.

3.4.2.5.6.1 Soldering temperature. Requirements herein and of the associated detail specification apply after terminals have been subject to a temperature of 240 °C for 10 seconds during soldering.

3.4.2.6 Optical connection. The designated connection (see 3.4.2.1) shall be as specified by the appropriate subparagraphs to this paragraph.

3.4.2.6.1 Pigtail cable.

3.4.2.6.1.1 Type. Optical fiber pigtail cables shall be in accordance with the requirements of MIL-C-85045/14. Additional buffering may be added to the optical fiber, to a maximum diameter of 900 µm, provided it does not cause either the optical fiber or pigtail cable to become non-compliant with their respective specifications.

3.4.2.6.1.2 Length. Pigtail cable length shall be measured from the point where it exits the case to the limit of the jacketed portion at the opposing end. Length shall be not less than one meter.

3.4.2.6.1.3 Mounting. Unless otherwise specified in the associated detail specification, the cable is to be perpendicular to the side of the case it exits. The pigtail cable shall be securely affixed to the package and the penetration sealed as required to provide the mechanical strength, hermeticity and functional performance specified herein.

3.4.2.6.1.4 Materials. The pigtail shall contain no metallic components exclusive of that portion of its length that is within 20 mm of the package.

3.4.2.6.1.5 Case junction strength. The package seal and pigtail cable/case junction shall retain their integrity and the device its capability to function as required herein and in the associated detail specification when subject to inspections, shipping, installation, use and maintenance performed in accordance with accepted procedures for such operations. The subparagraphs to this paragraph identify such stresses. Change in the value of the parameter identified in the associated detail specification, when the device is tested as specified in 4.7.4.1, shall be not greater than 0.4 dB while the stress is applied and 0.1 dB after it is removed.

3.4.2.6.1.5.1 Tension. The tensions specified in the subparagraphs to this paragraph apply for all angles less than 45 degrees with respect to the normal to the case at the point of attachment.

3.4.2.6.1.5.1.1 Fiber. To 5 N, applied to the fiber alone.

3.4.2.6.1.5.1.2 Sheath. To 10 N, applied to the sheath (including the strength member) alone.

3.4.2.6.1.5.2 Twist. Twists to 360 degrees per meter.

3.4.2.6.2 Connector.

3.4.2.6.2.1 Type. Connectors shall be receptacles and shall be as specified by the associated detail specification.

3.4.2.6.2.2 Mounting. The connector receptacle shall be an integral part of the device package.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the currentversion before use.

3.4.2.6.2.3 Case junction strength. The package seal and connector/case junction shall retain their integrity and the device its capability to function as required herein and in the associated detail specification when subject to inspections, shipping, installation, use and maintenance performed in accordance with accepted procedures for such operations. Functional requirements apply when the connector is subject to a bending moment of 0.01 N-m about any part of its junction with the case. Change in the value of the parameter identified in the associated detail specification, when the device is tested as specified in 4.7.4.2, shall be no greater than 0.4 dB while the stress is applied and 0.1 dB after it is removed.

3.4.2.7 Marking. Marking of the device package shall be in as specified in the subparagraphs to this subparagraph.

3.4.2.7.1 Quality. Marking quality shall be as specified:

a. Include all components identified by the subparagraphs to 3.4.2.7.3

b. All marking shall be legible when conformal coatings are used, in accordance with MIL-M-46508.

c. Resist eradication by solvents.

Laser marking of metal surfaces shall be subject to procuring activity approval.

3.4.2.7.2 Location. The Part or Identifying Number, inspection lot identification code, index point, electrostatic discharge sensitivity (ESDS) identifier, and warning (if applicable) shall be located on the exterior top surface of the device. Top surface refers to that which is opposite to the surface where the terminals penetrate the package. Other markings may also be located on the exterior top surface, if space available. If not, the bottom exterior surface may be used.

3.4.2.7.3 Markings. Specific markings to be included on the package shall be as specified in the subparagraphs to this paragraph.

3.4.2.7.3.1 Part or Identifying Number (PIN). See 6.7.

3.4.2.7.3.2 Inspection lot identification code. Devices shall be marked with a four digit code to identify the week of final seal. When read from left to right or top to bottom, the first two digits shall be the last two digits of the calendar year, and the last two digits shall be the week number of that year. If the number of a week is a single digit, it shall be preceded by a zero. A unique suffix letter shall be appended to the code to identify devices sealed in the same week but placed into different quality conformance inspection lots (see 4.5.2). The same suffix letter shall be used on all devices in a particular quality conformance inspection lot.

3.4.2.7.3.3 Index point. Devices shall be marked with a reference for lead identification or package orientation shall be provided. The symbol is left to the contractor's discretion.

3.4.2.7.3.4 Electrostatic discharge sensitivity (ESDS) identifier. Devices shall be marked with the appropriate ESDS identifier as specified:

ESDS Class	Marking	
1	Δ	(single equilateral triangle outline)
2	Δ Δ	(double equilateral triangle)
3		(no marking)

3.4.2.7.3.5 Warning. Transmitter devices shall be identified and marked in accordance with ANSI Z136.2-1988.

3.4.2.7.3.6 Contractor's identification. Devices shall be marked with the name or trademark of the contractor.

3.4.2.7.3.7 Contractor's symbol. Devices shall be marked with the contractor's designating symbol or CAGE code number.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the currentversion before use.

3.4.2.7.3.8 Other. Devices may contain other markings, at the contractor's discretion. These markings shall not obscure nor shall they resemble (e.g. no four or five digit codes) those that are required.

3.4.3 Internal construction.

3.4.3.1 Internal thin film conductors. Internal thin film conductors, if used, shall be in accordance with MIL-H-38534.

3.4.3.2 Internal lead wires. Internal lead wires shall be in accordance with MIL-H-38534.

3.4.3.3 Element to substrate bond shear strength. All active and passive elements shall have a substrate bond shear strength of (6.2×10^{-3} kg/mm^2) x (element attach area) or 2.5 kg, whichever is less.

3.4.3.4 Wire bond strength. The minimum allowable bond strength shall be 1.0 gram for .0254 mm diameter wire and 0.5 grams for wire with diameter less than .0254 mm inch (aluminum or gold).

3.4.4 Workmanship. Workmanship shall be in accordance with the production practices, workmanship instructions, and training aids identified in the contractor's quality assurance program (see 4.2). Soft handling practices and ESD controls shall be implemented.

3.5 Reliability. Reliability shall be as specified in the associated detail specification.

3.6 Maintainability. The devices specified herein are not maintainable.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspections requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.2 Responsibility for compliance. All devices shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection provisions in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.3 Responsibility for documentation. All documentation required by section 4 shall be available for procuring activity review.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the currentversion before use.

4.1.4 Certification of conformance traceability. Contractors or suppliers who offer the devices described by this specification shall provide written certification, signed by a corporate officer who has management responsibility for the production of the device, (1) that the device being supplied has been manufactured and tested in accordance with this specification and conforms to all of its requirements, (2) that all devices are as described on the certificate which accompanies the shipment, and (3) that the dealers and distributors have handled the devices in accordance with the requirements of JEDEC Publication 108, Distributor Requirements for Handling Electrostatic Discharge Sensitive (ESDS) Circuits. The responsible corporate official may authorize other individuals to sign the certificate. The certificate shall include the following information:

a. Contractor's name and address
b. Customer's or distributor's name and address
c. Device type
d. Lot date code
e. Quantity of devices in shipment from contractor
f. Statement certifying device conformance and traceability
g. Signature and date of transaction

4.1.5 Test equipment failure or operator error. Whenever a device is suspected to have failed as a result of faulty test equipment, operator error, or procedural faults, the failure shall be documented in the test record with all pertinent test information and shall be available for review by the procuring activity. If lot jeopardy (see 6.3.2.8) is involved, the procuring activity shall be immediately notified.

4.1.5.1 Disposition of devices. If test equipment or operator error is suspected as the root cause of the failure and the device has not been damaged or degraded, a replacement device from the same lot may be used to complete testing. The replacement part shall be subjected to all of the tests previously performed on the failed device prior to the continuance of its testing. The contractor, at his own risk, has the option of continuing with the test before the cause of the test equipment failure or operator error has been established. The lot or surviving portion of the lot may be resubmitted to the corrected screening tests.

4.1.6 Failure and corrective action. The contractor shall document all failures and take corrective action when trends (two or more of the same failure) occur.

4.1.7 Electrostatic discharge sensitivity (ESDS) classification. The ESDS classification of a device design shall be established as specified in 4.6.4. Production devices shall be marked (see 3.4.2.7.3.4) and handled in accordance with this classification. Prior to classification of a design type, any devices produced shall be handled in accordance with ESDS class 1.

4.1.8 Lead forming. Lead forming (e.g. bending) shall be performed prior to device screening (see 4.4).

4.2 Quality assurance program. The contractor shall have a quality assurance program which ensures that the design, manufacturing, inspection and testing complies with the quality assurance provisions herein and the applicable associated detail specification(s). New quality assurance programs which are initiated by the contractor for the purposes of meeting this provision shall comply with MIL-H-38534, Appendix A and 4.2.1 through 4.2.5 herein. A quality assurance program plan referred to herein as ("QA plan") shall be submitted for review and approval by the procuring activity.

4.2.1 Quality assurance personnel and areas. The QA plan shall list quality assurance personnel (by title), their functions, responsibilities and authority. The QA plan shall address the following subjects (as a minimum):

a. Personnel training

b. Design, material, and process change control

c. Electrostatic discharge handling control program

d. Soft handling procedures

e. Tool, gauge, and test equipment maintenance and calibration

f. Cleanliness and atmosphere control in work areas

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

g. Inspection of incoming materials and work in-process

h. Quality control operations

i. Quality assurance operations

j. Failure and defect analysis and data feedback

k. Corrective action and evaluation

l. Rework provisions

m. Incoming, in-process, and outgoing inventory control

4.2.2 Self audit program/manufacturing. The contractor shall establish a program to assess compliance with the quality assurance provisions contained herein and in the associated detail specification. The self audit program shall identify review areas, frequency of audits, and corrective actions to be taken when variations from approved procedures or requirements of this specification are identified.

4.2.3 Statistical process control (SPC) program. The use of statistical process control (SPC) is strongly encouraged. The SPC program shall include training, definition of controlled critical processes, installation of statistical control techniques and a control action system as defined by JESD19. Process control is recommended for wirebonding, lid seal, lead trim, and final lead finishing (solder dip, etc.). The implementing procedures shall identify frequency of reviews, sample size, reject criteria, allowable rework, and disposition of the failed device or lot. A procedure is required, in the event of process failure, to trace, recover, and dispose of all devices manufactured since the last successful test. The use of SPC control charts is recommended.

4.2.4 Manufacturing flow charts. A flow chart shall be developed to show all manufacturing processes, where all materials and subassemblies enter the flow and all manufacturing, inspection, testing and quality verification points. The chart shall identify all major documents used for inspection of materials and production processes, environments and controls.

4.2.5 Control and inspection of acquisition sources. The contractor shall maintain objective evidence that supplies and services used for manufacturing and testing conform to the requirements herein and in the associated detail specification. The contractor shall be responsible for adherence to the quality assurance program of manufacturing and testing which takes place outside the contractor's facility.

4.2.6 Execution of the quality assurance program. The contractor shall execute the quality assurance program while producing devices. The procuring activity shall have access to nonproprietary areas of the contractor's plant to verify compliance.

4.2.7 Change of quality assurance program. The contractor shall not implement any change in manufacturing, the quality assurance program, or the production line without a concurrent change in the quality assurance program documentation and approval from the procuring activity.

4.3 Manufacturing requirements. The contractor or associated subcontractor shall have production and test facilities needed to comply with the provisions herein and the associated detail specification.

4.3.1 Inspection and controls during manufacture. The contractor shall establish and maintain in-process production controls, quality controls, and inspections at appropriate points in the manufacturing process. These controls and inspections shall be in accordance with MIL-H-38534, Appendix A, and the associated detail specification.

4.3.2 Cleanliness and atmosphere control in work areas. Cleanliness and atmosphere in work areas where unsealed devices, or device components, are processed or assembled shall be controlled. All fabrication, assembly, and testing of the devices, prior to seal, shall be in class 100,000 environment as described in Federal Standard 209.

4.3.3 Device traceability . Productions travelers (see 6.3.2.15) or similar documentation shall be employed to record the complete manufacturing history, including rework (see 6.3.2.16), for each device of a production lot (see 6.3.2.13). Each device shall be uniquely identified for this purpose.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the currentversion before use.

4.3.4 Rework provisions. The need for rework (see 6.3.2.16) shall be identified and documented. Rework shall be performed in accordance with MIL-H-38534.

4.4 Device screening. Unless otherwise specified in the associated detail specification, all devices shall be subjected to the screens in table II, in the order listed. Internal visual shall be performed prior to sealing. Other screens shall be performed after the device has been sealed. Devices that fail any screen shall be rejected. Rework on rejects may be performed to the extent it is allowed as specified in 4.3.4, or rejects may be used for quality conformance inspection destructive tests, to the extent it is allowed as specified in 4.5.3. Rejects do not effect the remainder of the production lot.

4.4.1 Particle impact noise detection (PIND). Devices shall be subjected to the PIND screen (see 4.7.2.1), if required by the procuring activity (see 6.2). If so, the PIND screen shall be performed after the constant acceleration/mechanical shock and before burn-in.

TABLE II. Device screens.

Screen	Test paragraph
Internal visual	4.7.2.6
Stabilization bake	4.7.3.4
Temperature cycling	4.7.3.2
Mechanical shock/constant acceleration	4.7.3.5
Burn-in	4.7.3.1
Hermeticity (seal)	4.7.2.2
External visual	4.7.2.7

4.5 Quality conformance inspections (QCI).

4.5.1 General.

4.5.1.1 Inspections. Unless otherwise specified in the associated detail specification, the Group A, B, C, and D inspections described in 4.5.4, 4.5.5, 4.5.6, and 4.5.7, respectively, shall be performed on device inspection lots subsequent to the completion of the device screening of 4.4. Inspection lots and use of rejects for testing are described in 4.5.1.2 and 4.5.1.3, respectively.

4.5.1.2 Inspection failure. Devices which fail any test under any QCI inspection shall be rejected. Devices which fail any test under the Group A inspection do not affect the remainder of the inspection lot. If the number of failures exceeds that allowed for any test under the Group B, C, or D inspections, resubmission is allowed as specified in 4.5.1.3. If further failures occur under the resubmission provisions the entire inspection lot shall be rejected.

4.5.1.3 Resubmission. Failure analysis of parts failing Group B, C, or D inspections shall be performed and the failure mechanism determined and documented. Devices may not be resubmitted if the analysis indicates the failure is due to processing procedures, design faults or non screenable defects. Resubmission for Group A failure is not permitted. Resubmission for Group B, C, or D inspections is permitted at double the sample size with zero failures allowed.

4.5.2 Inspection lots.

4.5.2.1 Formation of lots. Devices shall be segregated into identifiable inspection lots (see 6.3.2.5). An inspection lot shall constitute a production lot (see 6.3.2.13).

4.5.2.2 Lot release. Production lots may be delivered after all applicable device screens and QCI have been successfully completed. Devices or production lots rejected as a result of a screen or QCI failure shall not be delivered.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the currentversion before use.

4.5.2.3 Disposition of samples. Device samples subjected to destructive tests (see 4.5.3.1) shall not be included in any delivered lot. Device samples subjected to non-destructive testing shall pass an external visual inspection performed as specified in 4.7.2.7, in order to be included in a delivered lot.

4.5.3 Use of rejects or partial assemblies for destructive tests. Rejects from device screening, Group A or Group B inspection tests may be used, with procuring activity approval, for the performance of destructive tests. Partially assembled devices may be used for tests which do not require completed devices, e.g. wire bond pull, where their quality is representative of that of a completed device. Destructive and non-destructive tests are identified in the subparagraphs to this paragraph.

4.5.3.1 Destructive tests. The following are considered destructive tests.

Bond strength
Moisture resistance
Lead integrity
Salt atmosphere
Die shear strength
Electrostatic discharge sensitivity classification test
Internal water vapor test

4.5.3.2 Non-destructive tests. The following are considered non-destructive tests.

Internal visual (precap)
Stabilization bake
Temperature Cycling
Mechanical shock/Constant Acceleration
Particle impact noise detection (PIND)
Burn-in
Hermeticity (seal)
External visual
Group A tests (see associated detail specification sheets)
Physical dimensions
Resistance to solvents
Solderability
Short circuit protection
Tension
Twist
Connector receptacle/case junction strength

4.5.4 Group A inspection. Group A inspections shall be as specified in associated detail specification.

4.5.5 Group B inspection. Group B inspections shall be as specified in table III. Group B inspection shall be performed on samples from each inspection lot, the quantity to be determined by the procuring activity (see 6.2).

4.5.6 Group C inspection. Group C inspections shall be as specified in table IV. Group C inspection shall be performed on samples from each inspection lot, the quantity to be determined by the procuring activity (see 6.2).

4.5.7 Group D inspection. Group D inspection comprises a steady state life test as specified in 4.7.3.3. and a reliability test as specified in 4.7.7. This inspection shall be performed on devices selected from the first inspection lot per contract, the quantity to be determined by the procuring activity (see 6.2). Procuring activities may waive this inspection for contractors who have performed and whose devices have passed this inspection on prior contracts, provided there have been no changes to the device design or production process.

4.5.7.1 Lot continuation. Devices that have been manufactured, and undergone screening and QCIs A through C, as specified herein, may be delivered during the Group D inspection (see 4.5.7). This is subject to procuring activity approval. In any case, devices shall not be delivered if the limit for Group D allowed failures (see 6.2) has been exceeded prior to or at the time of delivery.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

4.6 Production representative device (PRD; see 6.3.2.14) screens and inspections. When specified in the acquisition document (see 6.2) the screens and inspections, as specified in the subparagraphs to this paragraph, shall be performed on the specified number of devices (see 6.2).

4.6.1 Screens. The PRD screens specified in the associated detail specification.

4.6.2 QCI. The Group A (see 4.5.4) and Group B inspections (see 4.5.5).

4.6.3 Design performance inspection. The PRD performance inspection specified in the associated detail specification.

4.6.4 ESDS classification (see 4.1.7). Upon successful completion of the screening, QCI and design performance inspections, three devices of each design type in a particular contract shall be evaluated using the procedures of MIL-STD-883, Method 3015. The lowest rating of the three shall be designated ESDS classification for its particular design (see 3.4.2.7.3.4).

4.7 Test conditions and methods.

4.7.1 General. Provisions that apply to test methods specified herein are described in the subparagraphs to this paragraph. Test methods described in 4.7.2 through 4.7.6, and 4.7.8, shall be performed to the extent that they are specified by the device screening (see 4.4) and quality conformance inspection (see 4.5) provisions.

TABLE III. Group B inspections.

Parameter	Requirement paragraph	Test paragraph
Physical dimensions	3.4.2.1	4.7.2.8
Resistance to solvents	3.4.2.7.1	4.7.6.1
Solderability	3.4.2.5.6	4.7.2.5
Short circuit protection	3.2.3	4.7.5.1
Tension	3.4.2.6.1.5.1	4.7.4.1
Twist	3.4.2.6.1.5.2	4.7.4.1
Connector receptacle/case junction strength	3.4.2.6.2.3	4.7.4.2

TABLE IV. Group C inspections.

Parameter	Requirement paragraph	Test paragraph
Bond strength	3.4.3.4	4.7.3.6
Element shear strength	3.4.3.3	4.7.3.7
Internal water vapor content	3.4.2.3	4.7.2.3
Lead integrity	3.4.2.5	4.7.2.4
Moisture resistance	3.4.2.4.2	4.7.6.3
Salt atmosphere	3.4.2.4.2	4.7.6.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

4.7.1.1 Environment. Unless otherwise specified, the ambient temperature shall be 20 ± 5 °C and the relative humidity shall be between 20 percent and 80 percent.

4.7.1.2 Test equipment verification. Test equipment shall be maintained and calibrated in accordance with the test equipment manufacturer's recommendations. Calibration shall be traceable to National Institute of Standards and Technology (NIST) standards. Test equipment shall be evaluated daily to detect deviations from specified performance.

4.7.1.3 Test method deviation. Deviation from specified test methods requires procuring activity approval.

4.7.1.4 Contractor tests. Contractors performing tests other than described herein shall notify the procuring activity. The number of devices, test results, and test procedure shall be available for procuring activity review.

4.7.2 Package integrity test methods.

4.7.2.1 Particle impact noise detection. MIL-STD-883, method 2020, test condition A or B shall be performed.

4.7.2.2 Hermeticity. MIL-STD-883, method 1014 shall be performed. Fine leak testing shall be performed in accordance with conditions A or B and gross leak testing shall be performed in accordance with conditions C, D, or E. Consideration must be given to entrapment of gas within the optical connector or pigtail. Special test considerations and/or special testing conditions shall be invoked subject to procuring activity approval.

4.7.2.3 Internal water vapor content. Any of the three procedures of MIL-STD-883, Method 1018 may be used. The maximum allowable internal water vapor content at 100 °C shall be 5000 ppm.

4.7.2.4 Lead integrity. Electrical lead integrity shall be performed in accordance with MIL-STD-883, method 2004, condition B2 for leaded devices.

4.7.2.5 Solderability. MIL-STD-883, method 2003, shall be performed. The number of terminals to be tested shall be as specified in the acquisition document.

4.7.2.6 Internal visual. MIL-STD-883, method 2017 shall be performed. The fiber optic unique components internal to the package shall be inspected for compliance with the associated detail specification.

4.7.2.7 External visual. MIL-STD-883, method 2009, shall be performed. The fiber optic unique components of the external package shall be inspected for compliance with the associated detail specification.

4.7.2.8 Physical dimensions. MIL-STD-883, method 2016 shall be performed.

4.7.3 Stress test methods.

4.7.3.1 Burn-in. MIL-STD-883, method 1015 shall be performed. Conditions shall be as specified in the associated detail specification.

4.7.3.2 Temperature cycling. MIL-STD-883, method 1010, shall be performed. Conditions shall be as specified in the associated detail specification. External visual examination shall be as specified in 4.7.2.7.

4.7.3.3 Steady state life test method. MIL-STD-883, method 1005 shall be performed. Parameters and failure criteria shall be as specified in the associated detail specification.

4.7.3.4 Stabilization bake. MIL-STD-883, method 1008 shall be performed.

4.7.3.5 Mechanical shock/Constant acceleration. MIL-STD-883, method 2002 , test condition B or method 2001, test condition A shall be performed.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the currentversion before use.

4.7.3.6 Bond strength (see 6.3). MIL-STD-883, method 2011 shall be performed. The test condition shall be appropriate for the bond type, which shall be as specified in the production travelers. The device shall be preconditioned for one hour at +300°C minimum in either air or inert gas. At least one wire from each element and a minimum of 22 wires or the maximum number in the package, if less than 22, shall be pulled.

4.7.3.7 Element shear strength. MIL-STD-883, method 2019, shall be performed, on one of each element type within the package.

4.7.4 Optical connection test methods.

4.7.4.1 Pigtail cable/case junction strength. The method described in EIA-455-36 shall be performed on the fiber and jacket separately. Reference, therein, to 'connector', 'fiber optic cable' or 'cable' and 'transmittance' shall apply to the device case, pigtail cable, and loss respectively. The case shall be secured so that the stress is applied between the cable and case, only. Remove the jacket from 0.5 meter of the pigtail. Method 1 of EIA-455-6 shall be used to make attachment to the fiber and method 2 shall be used to attach to the jacket. The attachment must provide for applying both tension and twist. Applied tensions and twists shall be the maximum values specified in the subparagraphs to 3.4.2.6.1.5. The device shall be operated during the test. The parameter value specified in the associated detail specification shall be monitored. Devices will be considered as having passed if the value remains within the limit as specified in 3.4.2.6.1.5.

4.7.4.2 Connector/case junction strength. A special fixture for the package receptacle shall be constructed and fitted to the device under test. This fixture shall grasp the receptacle adjacent to the case without interfering with a mating cable connector and provide for applying a moment to the receptacle, alone, about its junction with the case. An optical fiber cable shall be connected to the device under test. The device shall be operated during the test. Measurements shall be made of the parameter value specified in the associated detail specification before, during and after the application of the maximum bending moment specified in 3.4.2.6.2.3. The moment shall be applied perpendicularly to the connector axis and parallel to the plane of the electrical terminal side of the package. Devices will be considered as having passed if the value remains within the limits specified in 3.4.2.6.2.3.

4.7.5 Electrical test methods.

4.7.5.1 Short circuit. MIL-STD-883, method 3011 shall be performed on the electrical terminals identified by the associated detail specification. For logic interfaces not listed in MIL-STD-883, method 3011, see the associated detail specification.

4.7.6 Chemical test methods.

4.7.6.1 Resistance to solvents. MIL-STD-883, method 2015 shall be performed.

4.7.6.2 Salt atmosphere. MIL-STD-883, method 1009 shall be performed.

4.7.6.3 Moisture resistance. MIL-STD-883, method 1004 shall be performed.

4.7.7 Reliability test method. MIL-STD-883, method 1016 shall be performed. Test parameters shall be as specified in the associated detail specification. The procuring activity will determine when and the number of times this test will be performed.

4.7.8 Functional test methods. Functional tests shall be as described in the associated detail specification.

5. PACKAGING (shipping and storage)

5.1 Packaging requirements. Packaging shall be in accordance with MIL-M-55565. In addition, these devices shall be packaged in conductive material or packaged in an antistatic material within an electrostatic shielding material. ESD packaging materials shall not be reused unless they have been thoroughly examined for cracks and other damage. DIP sticks/rails with an antistatic coating shall not be reused unless their antistatic coating has been tested and passed the resistivity requirements for antistatic material or better (i.e. dissipative or conductive).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

5.2 Marking of container. Devices shall be packaged as specified in 5.1 and the container marked in accordance with MIL-STD-1285 for the ESDS rating determined in 4.6.4. Other markings shall be included as applicable.

6. NOTES

(This section contains information of general or explanatory nature that may be helpful, but it is not mandatory.)

6.1 Intended use. Devices described by this specification and the associated detail specification convert between an electrical signal and an optical signal. An electrical signal, which encodes the information being transmitted, is input to the transmitter and output from the receiver. The optical signal, an incoherent carrier intensity modulated by the electrical signal waveform, is output from the transmitter and input to the receiver.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issues of the documents referenced (see 2.1).

c. Fabrication and evaluation of production representative devices (see 6.3.2.14). A requirement must be included in the solicitation that require offerors not producing acceptable devices within the two previous years to fabricate and evaluate no less than three production representative devices (PRDs) after a contract is awarded. PRDs may be produced separately or may be selected at random from an initial production lot. Evaluation consists of the inspection of 4.6. PRDs must pass the evaluation before production devices may be accepted under contract.

d. PRD and Group A inspections cite functional test procedures contained in Appendix B herein for verification of performance requirements. Contractors may have procedures equivalent to some or all of these. Solicitations which reference this specification should contain notification that manufacturers may use their own test procedures, subject to procuring activity approval. Such test procedures must be submitted for review in accordance with the applicable Data Item Description (DID) described in 6.2.1.

e. The inclusion or exclusion of the PIND screen, (see 4.4.1). The PIND screen generally is used to identify particle contamination within device packages intended for space applications. If the screen is to be included, the number of samples and the number of allowed failures per production lot must be identified (see 6.6 for default values). The PIN must be modified as specified in 6.7.

f. Sample sizes for quality conformance inspections. Paragraph 6.6 Contains default sample size/inspection lot. Procuring activities should explicitly identify those tables or alternatives in the acquisition documents.

6.2.1 Consideration of data requirements. The following data requirements should be considered when this specification is incorporated in a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a contract data requirements list (DD form 1423) must be prepared to obtain data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
4.1.4	DI-MISC-80678/T	Certification/data report	Note 1.
4.2	DI-QCIC-80107	Quality program plan	Note 2.
4.4 and 4.5	DI-NDTI-80809A/T	Test inspection/reports	Note 3.
6.2.D.	DI-NDTI-80603/T	Test procedure	Note 4.

Note 1: Only 10.1, 10.2, 10.2.4, 10.3.2, and 10.5 apply. For 10.3.2: "This certificate is to attest that all products being delivered are in accordance with contract requirements. It shall contain the information listed in MIL-M-24791, paragraph 4.1.4, A.- G."

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

Note 2: It is recognized that the establishment of a unique QA program for the purpose of this contract is not merited. Therefore, the contractor shall supply documentation with content as identified by this DID in the format that the contractor usually employs to present such information to customers. Inclusion of all items identified by this DID is not mandatory. The requirements of MIL-M-24791, paragraph 4.2, with respect to MIL-H-38534 are not mandatory.

Note 3: Delete 10.2.6.2, 10.2.6.3 and 10.2.7.3. Change the following paragraphs as indicated:

10.1.1: Second sentence, delete "capable of being..." Paper media shall be used. Cals format does not apply.

10.1.2: Add, "the test report shall contain transmitter and receiver information in separate sections, each section following the format described herein."

10.1.3: Delete last sentence.

10.2.1: Item e., substitute "enter title descriptive of contents." Add item "k. Distribution statement."

10.2.3.2: Item b., does not apply.

10.2.6.4: Add "test results shall indicate the test procedure was used in a manner that references descriptions in DI-NDTI-80603."

Note 4: Only 10.1, 10.2.1.1 (exclude b. and c.), 10.2.1.3, 10.2.2.2, 10.2.2.4 and 10.2.2.6 apply.

6.3 Definitions.

6.3.1 General. References for words and phrases used within this specification and its associated detail specification are provided as specified.

6.3.1.1 References. References which contain definitions for terms used herein and the associated detail specification are listed below. They are to be used for terms other than those contained in section 4 (see 6.3.1.2) or those used for pulse waveform descriptions (see 6.3.1.3). The order of precedence is 6.3.2 followed by the reference in the order listed.

a. EIA Standard EIA-440-A, Fiber Optic Terminology
b. Federal Standard 1037, Glossary of Telecommunication Terms
c. IEEE Standard Dictionary of Electrical and Electronics Terms
d. Webster's New Collegiate Dictionary

6.3.1.2 Section 4 definitions. References for terms used in section 4 and their order of precedence are paragraph 6.3.2 and MIL-H-38534 and documents referenced therein for definitions.

6.3.1.3 Pulse waveform descriptions. IEEE Std 194-1977 is first in precedence for definitions of terms used to describe pulse waveform characteristics.

6.3.2 Defined terms.

6.3.2.1 Binary intensity modulation. Intensity modulation where only two levels of the optical carrier are used to convey information.

6.3.2.2 Case. The case is that part of the device which encloses all circuit components and is penetrated only to provide electrical and optical connection to those components.

6.3.2.3 Case temperature. The highest temperature on the external surface of a device which is equaled or exceeded within an area that is a minimum of 1 percent of the device external surface area.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

6.3.2.4 Central wavelength. The mean wavelength of an optical source. The central wavelength of an LED is defined as the average of the half power wavelengths:

$$\lambda_c = \frac{(\lambda_1 + \lambda_2)}{2}$$

where:

λ_c = *Central wavelength of LED*
λ_1 = *Lower wavelength at half full power*
λ_2 = *Upper wavelength at half full power*

6.3.2.5 Inspection lot. See production lot.

6.3.2.6 Intensity modulation. Variation of the instantaneous power level of an optical carrier in proportion to the level of a modulating waveform.

6.3.2.7 Jitter. The short-term non-cumulative variations of the significant instants of a digital signal from their ideal positions in time. (CCITT vol X.1, Terms and Definitions) "Significant instants', as used herein and the associated detail specifications, refer to those times at which a digital signal waveform crosses a designated reference magnitude. The "ideal position" of each such instant is at a time displaced from the corresponding significant instant of a reference waveform by the amount equal to the delay between measured and reference waveforms. The value of that delay is the average of the delays associated with the significant instants of an unbiased pattern at the maximum modulation rate where the averaging time is greater than 0.1 second.

6.3.2.8 Lot jeopardy. A condition where the quality of an entire production lot is suspect.

6.3.2.9 Modulation rate. For digital modulation, the reciprocal of the unit interval measured in seconds. This rate is expressed in baud (variation FED-STD-1037A).

6.3.2.10 Optical interface. A circular aperture concentric with and normal to the axis of the pigtail fiber or connector, as appropriate, through which optical power flows, either into or out from the device. The spatial characteristics of the flow and the dimensions of the aperture are defined by the characteristics of the core of a reference fiber. The location of the aperture for a pigtail connection is at the end of the pigtail fiber and for a connector at the nominal location for the face of a mating terminus.

6.3.2.11 Optoelectronic propagation delay. The average time required for a transition to propagate from an input to an output of a device. This time is measured between the mesial points of the associated transition waveforms. The reference magnitude for an electrical logic interface is that of the associated logic family. The reference magnitude for the optical signal is at 50% reference magnitude for the modulation envelope waveform.

6.3.2.12 Power spectral density. The power spectral density is a mathematical expression which describes the variation with wavelength of the emitted power from an LED or LD. It is defined at a wavelength to be the limiting value of the ratio of the emitted power in an interval including the wavelength to the width of the interval as the width approaches zero (0).

6.3.2.12.1 LED. Power spectral density for LEDs is to be based on a wavelength interval between 3 % and 9% of the required emission spectral width as opposed to the zero (0) limit cited above.

6.3.2.13 Production lot. A production lot consists of devices manufactured on the same production line using the same production techniques, materials, controls, and design. This specification does not distinguish between a production lot and an inspection lot. Devices constituting a lot are sealed within a six week period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

6.3.2.14 Production representative device. A production representative device (PRD) is a device which satisfies all of the requirements of the general specification and an associated detail specification. It is to be used for the evaluation of design, performance and production potential. It is constructed in an identical way to that which would be produced on a production line.

6.3.2.15 Production travelers. The paperwork associated with each device which accompanies the device lot as it is processed through the manufacturing cycle.

6.3.2.16 Rework. The process by which a device is brought into conformance.

6.3.2.17 Spectral width. A measure of the wavelength extent of an optical source.

6.3.2.17.1 LED. The fullwidth-half maximum (FWHM) spectral width of an LED is defined as the full width of the wavelength difference at half maximum power spectral density and can be calculated as follows:

$$\Delta\lambda_{FWHM} = \lambda_2 - \lambda_1$$

where:

$\Delta\lambda_{FWHM}$ = *Spectral width of LED*
λ_1 = *Short wavelength at half full power*
λ_2 = *Long wavelength at half full power*

6.4 Abbreviations. Abbreviations for terms or phrases used in this specification are listed below.

a. ANSI: American National Standards Institute
b. DID: Data Item Description
c. DOD: Department of Defense
c. DODISS: Department of Defense Index of Specifications and Standards
e. DUT: Device under test.
f. EIA: Electronic Industries Association
g. ECL: Emitter Coupled Diode
h. ESD: Electro-Static Discharge
i. ESDS: Electro-Static Discharge Sensitivity
j. FWHM: Full Width Half Maximum
k. IEEE: Institute of Electrical and Electronic Engineers
l. LD: Laser Diode
m. LED: Light emitting diode.
n. PIND: Particle Impact Noise Detection
o. PRD: Production Representative Device
p. QA: Quality Assurance
q. QCI: Quality Conformance Inspection
r. SPC: Statistical Process Control

6.5 Symbols.

6.5.1 References. The following documents are references for symbols used in this specification. Unless specified otherwise, the order of precedence shall be the order listed.

1. ANSI/IEEE Std 280-1985, IEEE Standard Letter Symbols for Quantities Used in Electrical Science and Electrical Engineering

2. ANSI/IEEE Std 260-1978, IEEE Standard Letter Symbols for Units of Measurement

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

6.5.2 Symbols definitions.

6.5.2.1 V_{CC}. Designates the most positive power supply voltage.

6.5.2.2 V_{EE}. Designates the most negative power supply voltage.

6.6 QCI sampling and accept/reject criteria.

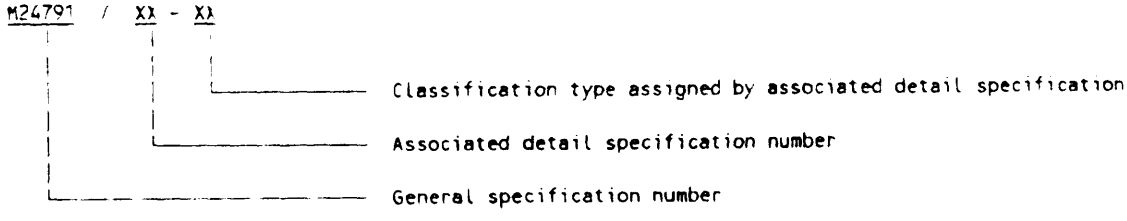
6.6.1 Sample sizes and accept/reject criteria. Tables V specifies sample sizes and the number of allowed failures per inspection lot for QCI for a lot size of 50. Acquisition documentation must reference this table or provide an alternative.

TABLE V. Sample sizes and failure criteria for QCI routine 1 1/.

Parameter	Test paragraph	Sample size/ inspection lot	Allowed failures/ inspection lot
Physical dimensions	4.7.2.8	2	0
Resistance to solvents	4.7.6.1	2	0
Solderability	4.7.2.5	1 (all leads)	0
Short circuit protection	4.7.5	1	0
Tension	4.7.4.1	3 or 5	0 or 1
Twist	4.7.4.1	3 or 5	0 or 1
Connector/case junction strength	4.7.4.2	3 or 5	0 or 1
Bond strength	4.7.3.6	1 (all bonds)	0
Die shear strength	4.7.3.7	1 (2 of each element type)	0
Internal water vapor content	4.7.2.3	3 or 5	0 or 1
Lead integrity	4.7.2.4	1 (all leads)	0
Moisture resistance	4.7.6.3	3 or 5	0 or 1
Salt atmosphere	4.7.6.2	3 or 5	0 or 1
Steady state life test	4.7.3.3	5 2/	0
Reliability test	4.7.7	5 2/	0

1/ An "or" option in the table implies that only the lesser number of samples need be tested with the lesser number of failures applying; if failures occur, the sample size is increased to the greater number with the greater number of failures becoming the new limit.
2/ From the first inspection lot (see 4.5.7).

6.7 PIN. The Part Identifying Number (PIN) (see 3.4.2.7.3.1) shall be as follows:



If PIND screening is included the suffix "PIND" shall be attached to the part number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

6.8 Subject term (key word) listing.

Bit error rate
Hermetic
Hybrid microcircuit
Jitter
Light Emitting Diode (LED)
Modulation
Optical
Optical interface
Optical pigtail
Optical signal
Optoelectronic

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX A

DEFINITIONS FOR LOGIC INTERFACES

10. SCOPE

10.1 Scope. This appendix defines input and output interfaces for emitter coupled logic (ECL) families. These definitions identify those characteristics required of digital fiber optic device electrical interfaces intended to interface with those families. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. LOGIC FAMILY INTERFACE DESCRIPTIONS.

30.1 Emitter coupled logic.

30.1.1 Logic levels. Logic levels are stated with respect to the V_{CC} supply (see 6.5.2.1).

30.1.2 Input.

30.1.2.1 Logic HIGH.

30.1.2.1.1 Voltage level range. Input levels greater than -1.2 V and less than -0.75 V are defined to logic HIGH.

30.1.2.1.2 Maximum current. The maximum current into a device input terminal shall be no greater than 400 µA at any allowed logic HIGH voltage level.

30.1.2.2 Logic LOW.

30.1.2.2.1 Voltage level range. Input levels less than -1.5 V and greater than -1.8 V are defined to be logic LOW.

30.1.2.2.2 Maximum current. The maximum current out of a device input terminal shall be no greater than 10 µA at any allowed logic LOW voltage level.

30.1.2.3 Transition duration. Transfer characteristics apply for input transition durations to 2.5 ns (see 6.3.1.3) between proximal and distal levels at 20 percent and 80 percent of pulse amplitude range.

30.1.2.4 Input capacitance. Input capacitance shall be no greater than 7 pF.

30.1.3 Output.

30.1.3.1 Logic HIGH level. Output levels greater than -1.0 V and less -0.75 V are defined to be a logic HIGH.

30.1.3.2 Logic LOW level. Output levels less than -1.65 V and greater than -1.8 V are defined to be a logic LOW.

30.1.3.3 Transition duration. Output transition durations shall not exceed 2 ns, between proximal and distal levels at 20 percent and 80 percent of pulse amplitude.

30.1.3.4 Drive capability. Output levels defined above apply for a load consisting of 50 ohms connected to -2.0 V.

30.1.4 Reference magnitude. The reference magnitude for temporal measurements with ECL signals is -1.38 V with respect to V_{CC}.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B

TEST METHODS

10. Scope.

10.1 Scope. This appendix contains test methods to be used for evaluating digital fiber optic devices. They are intended to be referenced for use and usually require that the referencing document supply the evaluation criteria and sometimes, test conditions. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS.

20.1 Government documents.

20.1.1 Standard. The following standard forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue of this document is that listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation, (see 6.2).

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics

20.2 Non-government publications. The following document forms a part of this specification to the extent specified herein. The issue of this document is indicated in the solicitation. Issues of DOD adopted documents are those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited the solicitation (see 6.2).

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

IEEE Std 194-1977 - Pulse Terms and Definitions

(Copies of IEEE publications are available from Global Engineering Documents, 2805 McGaw Avenue, Irvine, CA, 92714.)

30. Test Methods. Test methods for functional tests for digital fiber optic transmitters and receivers are described in subparagraphs 30.1 through 30.12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B

30.1 **CONDITIONS AND PROCEDURES APPLICABLE TO ALL TEST METHODS.**

30.1.1 Scope. This test method contains material which is common to all test methods.

30.1.2 Pulse waveform terminology. See IEEE Std 194-1977, Pulse Terms and Definitions, for definitions of terminology used to describe pulse waveforms.

30.1.3 Device operation.

30.1.3.1 Reference environment. The reference environment for performance tests shall have an ambient temperature within the range of 15 to 25 °C and humidity between 20 and 80 percent. Unless otherwise specified by the requiring document, temperature shall vary no more than ± 1 °C during a test.

30.1.3.2 Test jig. A test jig is required to perform all powered performance tests. This jig must provide:

(1) a mechanically secure platform to support the device under test (DUT)
(2) temporary connections to the electrical pins
(3) the electrical and thermal environment for the device to function as intended in normal application and
(4) a means to apply stimulus to and receive response from the DUT.

Special jigs may be required if the device is to be powered during mechanical (shock and vibration) and/or environmental (temperature and humidity) testing. Unless otherwise specified, heat transfer shall be solely through natural convection, radiation and conduction representative of normal application.

30.1.3.3 Orientation.

30.1.3.4 Insertion of DUT into test jig. The device under test (DUT) must be placed into the test jig after all the equipment is set to perform the test. It is not good practice to insert the device into a "live" receptacle. Therefore the test arrangement should provide for grounding all terminal sockets until the device is fully inserted. This detail should be observed for each test method but will not be explicitly stated therein.

30.1.3.5 Power supply voltages. When a test method is referenced the operating voltage(s) to be used for testing should be identified. If not identified, the nominal value shall be used. Tests may be repeated with other voltages within the required range. Correct power supply connections and verification of proper operating voltages are required by all test methods even if not stated explicitly.

30.1.4 Test equipment.

30.1.4.1 Required.

30.1.4.1.1 Test method listing. Each test method includes a listing of required test equipment. Capabilities identified in that listing are in addition to those required by this test method.

30.1.4.1.2 Substitution. Substitutions may be made for the "required test equipment" (and to the degree that this affects procedure, procedure also) provided that such substitutions meet the critical performance parameters for the associated recommended test equipment. The essential test method shall remain the same unless it is established that a proposed alternate method provides equivalent or greater assurance that requirements are being met.

30.1.4.2 Calibration. The calibration of all test equipment used for tests shall be current and traceable to the appropriate NIST standards.

30.1.4.3 Settings. Equipment required to perform tests must be adjusted to set the required operating conditions. These adjustments will be identified but detailed instructions will not be provided.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

30.1.4.3.1 Oscilloscope settings. Unless otherwise specified, when it is indicated that the oscilloscope controls be set to display a waveform, the objective shall be to include one full period or the maximum extent of the particular waveform feature in the horizontal extent of the display and to obtain the maximum on screen vertical extent. Unless otherwise specified, when a synchronizing signal is present the oscilloscope shall be triggered from that signal.

30.1.5 Setup and test connections. Each test requires that the test jig be properly positioned and all necessary equipment be connected to it. Such equipment includes power supplies, stimulus sources and response monitors. Test methods will not explicitly state these necessary operations. Test arrangement drawings will be provided to indicate the arrangement of DUT, jig, and test equipment.

30.1.5.1 Signal lines. Unless otherwise specified, all signal cables shall be 50 ohm coax and all instrument inputs terminated with 50 ohms. Only line lengths having negligible effect on waveforms shall be used.

30.1.6 Procedure. The performance of each test procedure shall include a delay between the application of power supply voltage and the application of test signals. Another delay shall be allowed between the application of test signals and any measurement to allow the device to reach a stable operating condition. This "warm-up" time should be identified in the requiring document. If it is not, an appropriate time will need to be determined. Devices utilizing more than one power supply voltage may require that they be applied in a specific sequence. Test signal characteristics should be established before being applied to the DUT.

Sets of measurements correlated by time or temperature should be performed in an expeditious manner. This is true of virtually all related measurements. For this reason, the use of automated techniques is encouraged.

30.1.7 Test record. The performance of a test method includes recording all pertinent data, including a detailed description of the method used (which may be accomplished through reference), environmental information (actual ambient temperature and humidity of the test environment) and measured results, if not explicitly stated therein. Details of jig construction shall be included in the data.

30.1.8 Tolerances. Tolerances of MIL-STD-883 shall apply.

30.1.8.1 Measurement errors. Requirements are stated as absolute quantities and do not include measurement errors. Requirements state minimum acceptable performance. Consequently, either measured data or acceptance criteria must be adjusted for measurement errors. For example, measurement uncertainty errors must be either added to or subtracted from the required parameter values, as appropriate, to establish pass/fail criteria which assure the highest probability that a requirement is being met.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the currentversion before use.

30.2 **POWER SUPPLY VOLTAGE AND CURRENT**

30.2.1 Scope. This test method describes how the device under test (DUT) is to be powered during other tests and the procedure used to determine device current requirements at a particular operating voltage.

30.2.2 Symbols used.

DUT: device under test

30.2.3 Test equipment.

30.2.3.1 Required. Test equipment required by this test method is listed in table VI.

Table VI. Required test equipment.

Item	Generic name	Capabilities	Suggested Mfr/Model
01	DUT test jig	See 30.1, 30.3, 30.4	
02	Power supply(ies)	Remote sense; current limit; off line set	
03	Meters		
04	Stimulus and loads	As required	

30.2.4 Setup. The considerations and procedures of 30.1 shall be observed or executed, as appropriate. Power supply connections are illustrated in figure 1. Specifications apply to power supply voltages at the device electrical terminals. Consequently, four wire, or Kelvin, connections shall be employed to control power supply voltages. Separate leads shall be used for voltage measurement. The device under test (DUT) shall be operated as specified in 30.1 and 30.3 or 30.4, as appropriate, and by the referencing document. All input signals and loads required to produce the required operating conditions must be present. Only the current specified shall be measured; the current delivered to loads is specifically excluded for some devices.

30.2.5 Procedures.

30.2.5.1 Measurement of power supply voltage and current. Apply the required operating conditions. Set the power supply voltage(s) as required. Measure and note the required operating current(s) associated with each power supply voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

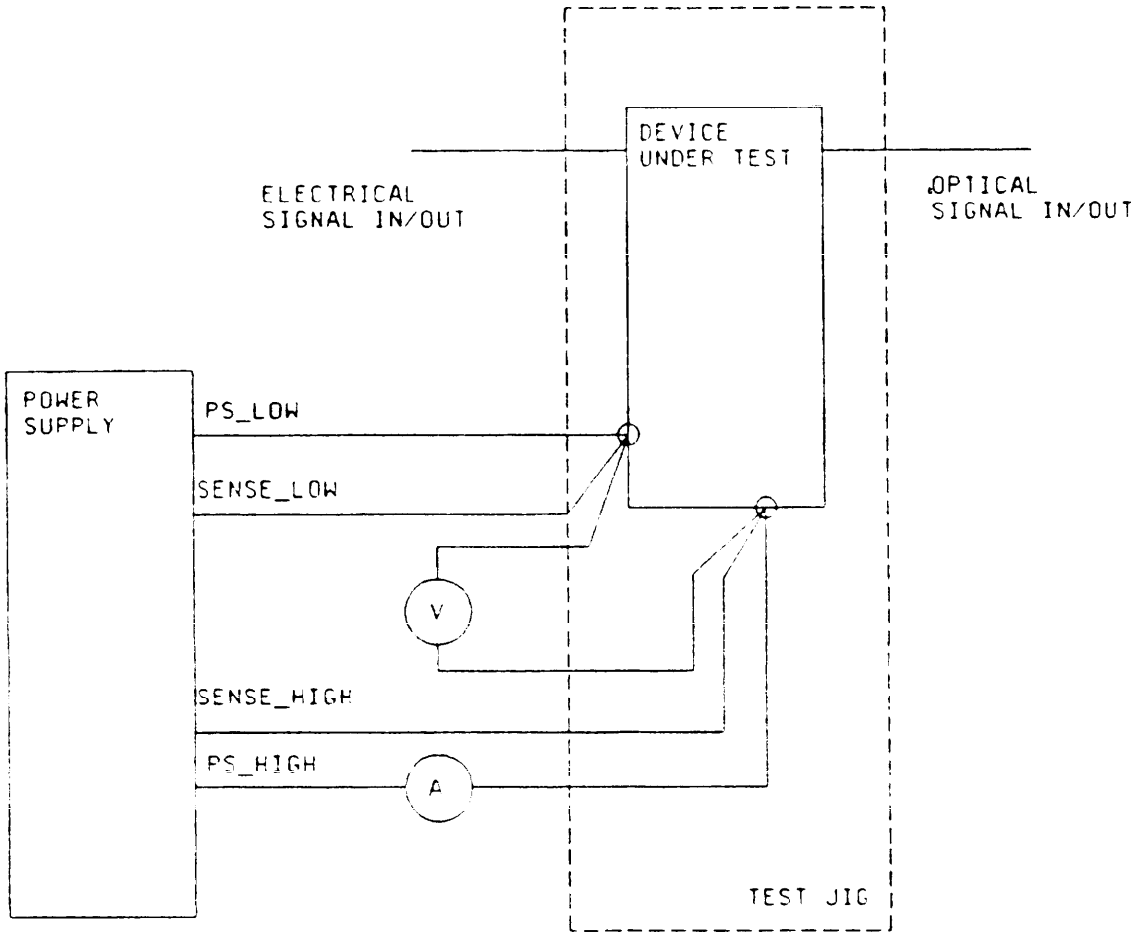


FIGURE 1. Power supply connections.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T02:42Z
Check the source to verify that this is the current version before use.

30.3 **BASIC SETUP FOR DIGITAL TRANSMITTER TESTS**

30.3.1 Scope. Many digital transmitter tests employ setups that are similar. To avoid repetition, this test method describes required equipment, interconnections and considerations common to a number of transmitter tests. It is to be referenced by those test methods.

30.3.2 Symbols used.

DUT: Device under test
L_s: Sum of optical fiber coupler coupling ratio (dB) and setup loss
MR_L: Lowest modulation rate required of the DUT
MR_H: Highest modulation rate required of the DUT
P_r: Value of optical power meter indication during procedure for establishing the value of L_s
P_m: Value of power meter indication during performance of a test method procedure
P_{out}: Value of optical power at the output from the transmitter under test

30.3.3 Test equipment.

30.3.3.1 Required. Test equipment required by this test method is that listed for the "power supply voltage and current" test method (see 30.2) and in table VII. The capabilities required of test equipment associated with the referenced test method may differ from those identified herein (see table VII and 30.3.3.2).

Table VII. Required test equipment.

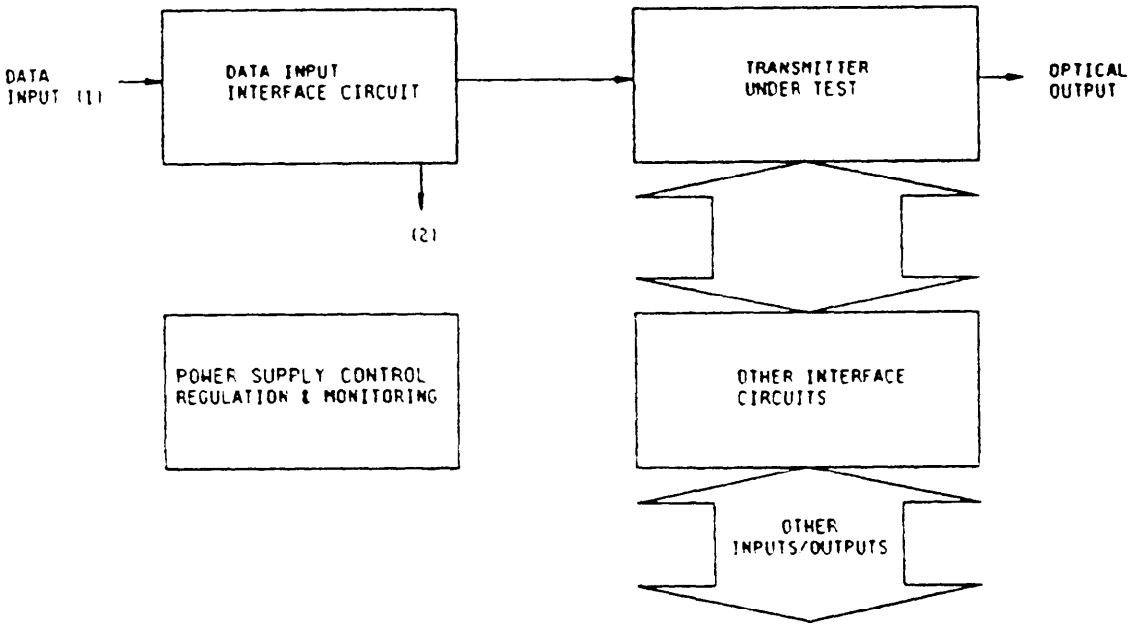
Item	Generic name	Capabilities	Suggested Mfr/Model
01	clock generator		
02	electrical signal splitter (2)		
03	pattern generator		
04	jitter modulator	see 30.3.3.4 herein	
05	transmitter test jig	see 30.3.3.3 herein	
06	optical clad mode stripper		
07	optical fiber coupler	90% and 10% outputs	
08	optical power meter		
09	optical to electrical (O/E) converter	analog; dc response	
10	electrical to optical (E/O) converter	binary intensity modulated	
11	measurement equipment		

30.3.3.2 Considerations. The clock generator supplies the timing and the pattern generator the data waveform required and described by the referencing test method. The combination of clock and pattern generators shall be capable of producing a square wave pattern with duty factor (DF) of 0.5 ± 0.05 and jitter (of the waveform when displayed on an oscilloscope) less than 20 ps rms. Details are peculiar to specific test methods. The jitter modulator (see 30.3.3.4) should produce negligible output jitter when receiving no modulation input. Electrical signal splitters provide samples of test signals to measuring instruments. The cladding mode stripper insures that only core modes are measured. The optical power meter and optical to electrical (O/E) converter shall be appropriate to the measurements to be made with respect to all critical parameters including wavelength, input power levels, sensitivity, accuracy, stability, linearity, etc. The optical to electrical (O/E) converter converts the modulation envelope waveform of the optical signal to a proportional electrical signal for measurement. Its contribution to waveform measurement error shall not exceed 10%. See 30.4.3.3 for a discussion of the E/O converter.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B

30.3.3.3 Test jig. Minimum requirements for the test jig are given in 30.1. The test jig provides an electrical signal environment which includes the functions identified in figure 2. The interface circuit(s) to the transmitter data input(s) provide the waveform, levels and impedance required by the transmitter under test, properly terminate the signal source supplying the test jig data input and provide any synchronizing or reference signals required by the measurement equipment. Interface circuits at other transmitter inputs (or outputs) perform similar functions. With respect to this method and any referencing method, the input and output interfaces of the test jig are assumed to be 50 ohms to ground, direct coupled. Delay between monitor outputs for signals applied to the DUT and the signals at the pins of the DUT shall be minimum. Bypassing may be provided at the power supply pins of the DUT to the extent allowed by the requiring document.



NOTES:
1. Single ended or differential.
2. Optical carrier modulating waveform sample (differential).

FIGURE 2. Transmitter test jig.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

30.3.3.4 Jitter modulator. The effect of the jitter modulator on the transitions of the signal it receives from the pattern generator is similar to the jitter on transitions at the digital output of a fiber optic receiver. Figure 3 is a block diagram for a jitter modulator. The carrier input is direct coupled through to the comparator input. The modulation input is ac coupled to the summing circuit. The summing circuit must isolate the carrier and modulation signal sources to minimize their interaction. The comparator threshold is set to a level appropriate for the carrier signal. Output jitter is assumed to be negligible when no signal is applied to the modulation input.

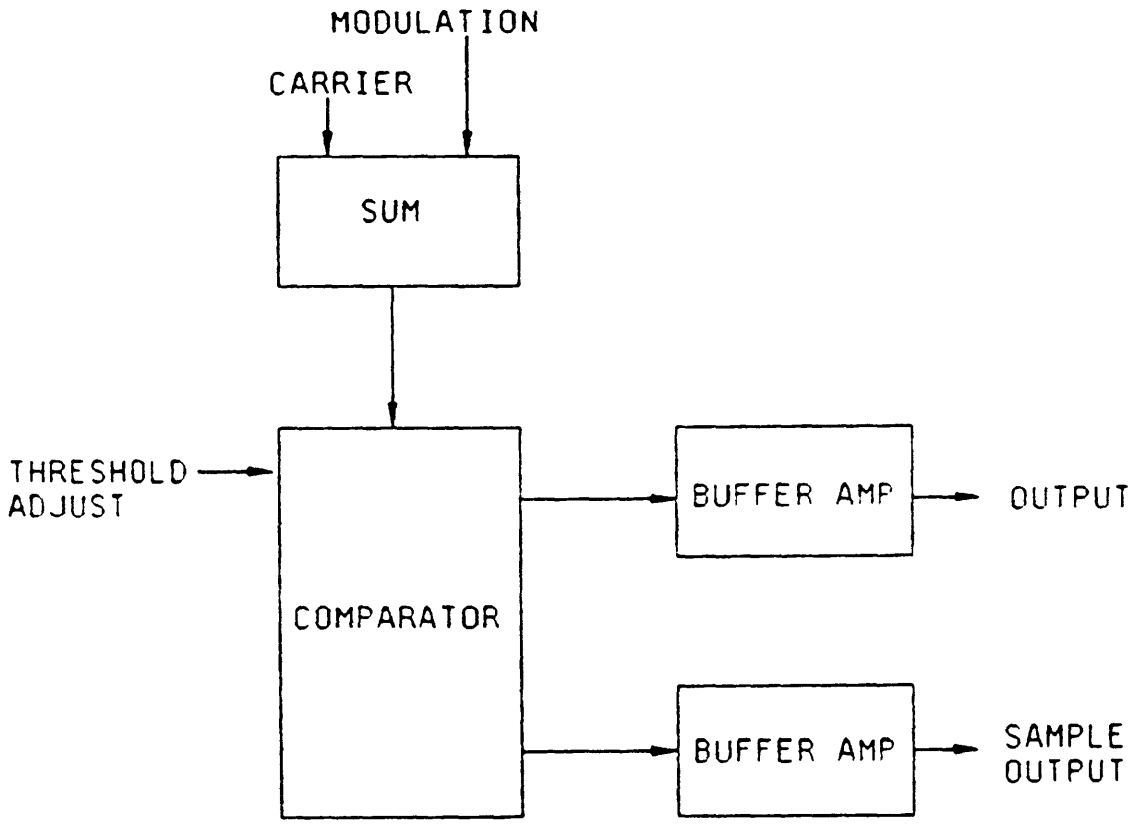
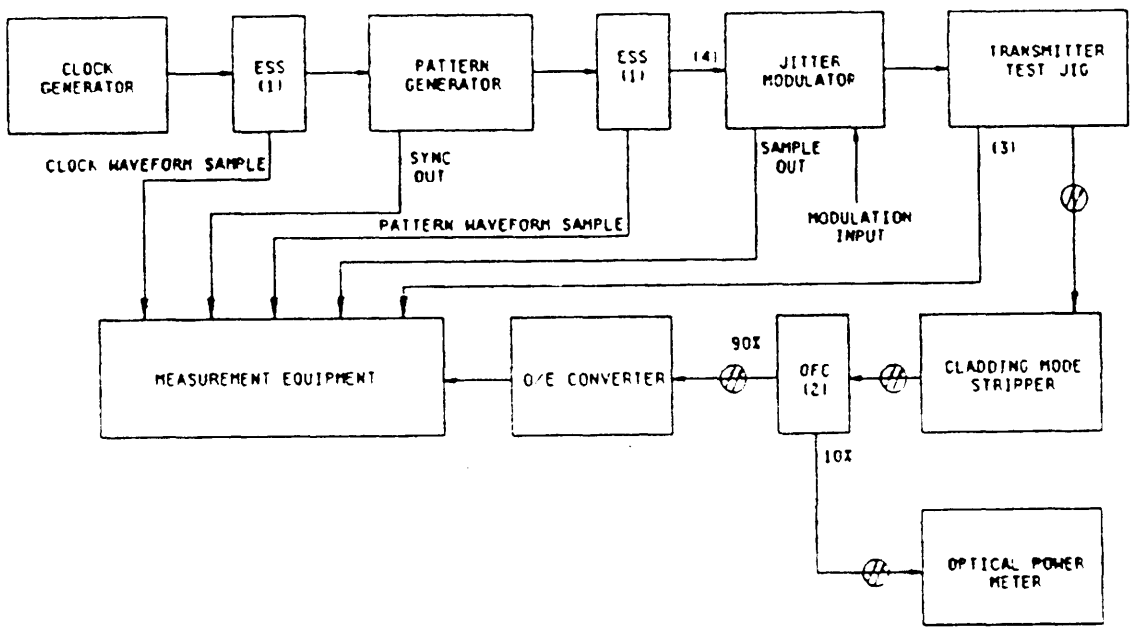


FIGURE 3. Jitter modulator.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

30.3.4 Setup. The considerations and procedures of 30.1 shall be observed or executed, as appropriate. The transmitter under test is to be powered as described by the "power supply voltage and current" test method (see 30.2). The interconnection of the required test equipment is illustrated in figure 4.



NOTES:
1. ESS = Electrical signal splitter.
2. OFC = Optical fiber coupler, 10% and 90% outputs.
3. Optical carrier modulating waveform sample (differential).
4. Carrier input.

FIGURE 4. Basic setup for digital transmitter tests.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

30.3.5 Procedures.

30.3.5.1 Optical power measurement.

30.3.5.1.1 Establishing a reference. The power at the transmitter output may be computed from the optical power meter indication, the coupling ratio (dB) of the optical fiber coupler and the setup loss. This procedure determines the combination of coupling ratio and setup loss, designated herein as L_s(dB). Unless otherwise specified by the referencing document, the pattern is a square wave and the modulation rate is $(MR_H+9*MR_L)/10$ where MR_H and MR_L are, respectively, the lowest and highest required modulation rates for the DUT. If run length is specified separately, MR_L shall be the lowest required modulation rate divided by the run length. Measure and record the power out of the 10% port of the optical fiber coupler, P_r (dBm). Connect the power meter in place of the input to the optical fiber coupler. Measure and record this power, P_{out}(dBm). The difference, P_{out}(dBm)-P_r(dBm), equals L_s(dB). Return the power meter to the coupler 10% output position.

30.3.5.1.2 Making a measurement. The output power from a DUT may be determined while performing any test method procedure without doing a direct measurement. Note the power meter indication, P_m(dBm). P_{out}(dBm), the average transmitter output power level, may be calculated as follows:

$$P_{out}(dBm) = P_m(dBm)+L_s(dB)$$

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B

30.4 **BASIC SETUP FOR DIGITAL RECEIVER TESTS**

30.4.1 Scope. Many digital receiver tests employ setups that are similar. To avoid repetition, this test method describes required equipment, interconnections and considerations common to a number of receiver tests. It is to be referenced by those test methods.

30.4.2 Symbols used.

A: Value of attenuator setting during performance of a test method procedure
DUT: Device under test
L_s: Sum of coupling ratio (dB) and setup loss
MR_L: Lowest modulation rate required of the DUT
MR_H: Highest modulation rate required of the DUT
P_r: Value of optical power meter indication during procedure for establishing the value of L_s
P_m: Value of power meter indication during performance of a test method procedure
P_{in}: Value of optical power at the input to the receiver under test

30.4.3 Test equipment.

30.4.3.1 Required. Test equipment required by this test method is that listed for the "power supply voltage and current" test method (see 30.2) and in table VIII. The capabilities required of test equipment associated with the referenced test method may differ from those identified herein (see table VIII and 30.4.3.2).

30.4.3.2 General considerations. The clock generator supplies the timing and the pattern generator the waveform as required and described by the referencing test method. The combination of clock and pattern generators shall be capable of producing a square wave pattern with duty factor (DF) of 0.5 ± 0.05 and jitter (of the waveform when displayed on an oscilloscope) less than 20 ps rms. Details are peculiar to specific test methods. The jitter modulator (see 30.3.3.4) should produce negligible output jitter when receiving no modulation input. Electrical signal splitters (ESS) are used to provide samples of test signals to measuring instruments. The optical fiber coupler provides a means to monitor the power delivered to the receiver under test. The power meter operates with an essentially constant input so that its measurement error is constant. Input power to the receiver is varied by adjusting the optical attenuator. The cladding mode stripper insures that only core modes are used to test the receiver. The optical to electrical (O/E) converter converts the modulation envelope waveform of the optical signal to a proportional electrical signal for measurement. Its contribution to waveform measurement error shall not exceed 10%.

Table VIII. Required test equipment.

Item	Generic name	Capabilities	Suggested Mfr/Model
01	clock generator		
02	electrical signal splitter (2)	equal output levels	
03	pattern generator		
04	jitter modulator	see 30.3	
05	electrical to optical (E/O) converter	binary intensity modulated	
06	optical to electrical (O/E) converter	analog; dc response	
07	optical fiber coupler (2)	10%:90% ratio; excess loss less than 0.5 dB	
08	optical power meter		
09	optical to electrical (O/E) converter		Hewlett-Packard model
10	optical attenuator	calibrated	8153A with 81531A
11	optical clad mode stripper		power sensor
12	receiver test jig	see 30.4.3.3.3 herein	
13	measurement equipment		

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the current version before use.

30.4.3.3 E/O converter.

30.4.3.3.1 Function. The electrical to optical (E/O) converter produces a binary intensity modulated optical output signal with the required optical characteristics (average wavelength, spectral width, power, extinction ratio, etc.). The modulation envelope waveform of the optical output signal shall maintain the timing relationships (pulse widths and transition jitter) of the input signal sufficient for the measurement being performed. Envelope waveform transition durations shall be adjustable.

30.4.3.3.2 Extinction ratio. The extinction ratio of the output shall be adjustable. Unless otherwise specified by the referencing document, optical signal modulation extinction ratio shall be 10 +2/-0. If the required extinction ratio is outside the adjustment range, the need for correction should be considered.

30.4.3.3.3 Test jig. Minimum requirements for the test jig are given in 30.1. The test jig provides an electrical signal environment which includes the functions identified in figure 5. Interface circuits at the receiver data output(s) provide for proper termination of these ports and for delivering accurate samples of their output signals to measuring equipment. Interface circuits at other receiver outputs (or inputs) perform a similar function. With respect to this method and any referencing method, the input and output interfaces of the test jig are assumed to be 50 ohms to ground, direct coupled. Delay between monitor outputs for signals applied to the DUT and the signals at the pins of the DUT shall be minimum. Bypassing may be provided at the power supply pins of the DUT to the extent allowed by the requiring document.

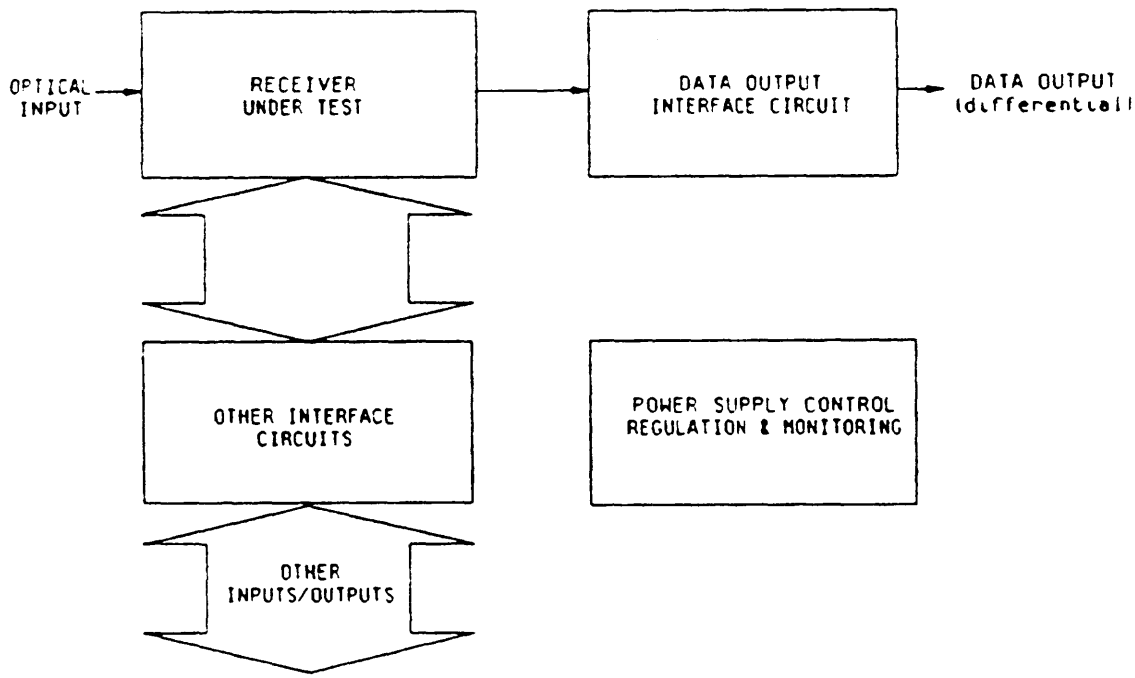
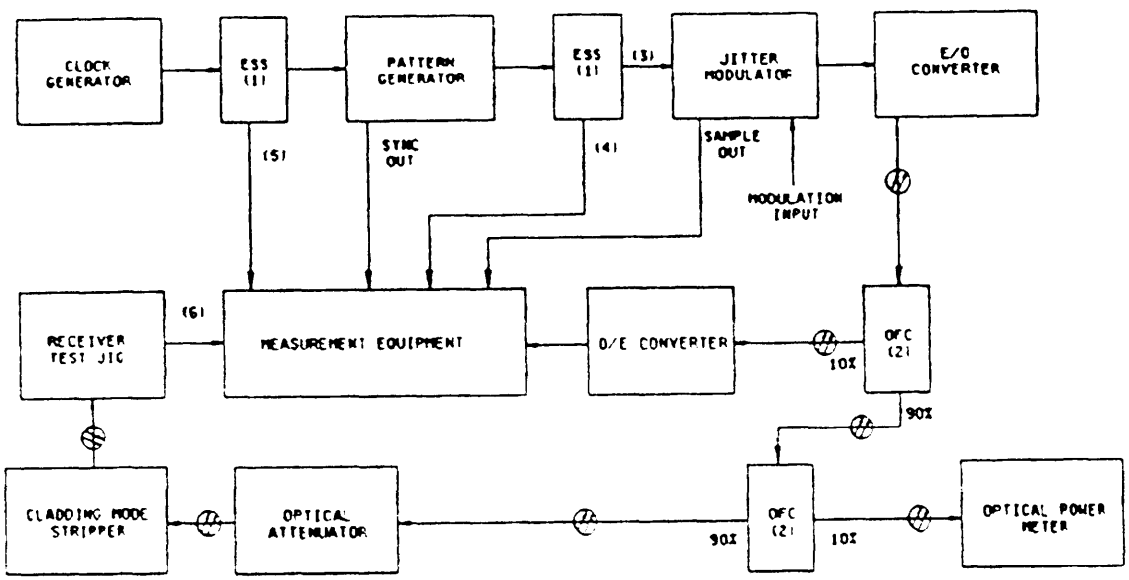


Figure 5. Receiver test jig.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B

30.4.4 Setup. The considerations and procedures of 30.1 shall be observed or executed, as appropriate. The receiver under test is to be powered as described in the "power supply voltage and current" test method (see 30.2). The interconnection of the required test equipment is illustrated in figure 6.



NOTES.
1. ESS = electrical signal splitter.
2. OFC = optical fiber coupler, 10% and 90% outputs.
3. Carrier input.
4. Pattern waveform sample.
5. Clock waveform sample.
6. Data output from receiver test jig.

Figure 6. Basic setup for digital receiver tests.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

30.4.5 Procedures.

30.4.5.1 Optical power measurement.

30.4.5.1.1 Establishing a reference. The power delivered to the receiver input may be computed from the optical power meter indication, the coupling ratio (dB) of the optical fiber coupler, the attenuator setting and setup loss. This procedure determines the combination of coupling ratio and setup loss designated herein as L_s(dB). Unless otherwise specified by the referencing document, the pattern is to be a square wave at a modulation rate of $(MR_H+9*MR_L)/10$ where MR_L and MR_H are, respectively, the lowest and highest required modulation rates for the DUT. If run length is specified separately, MR_L shall be the lowest required modulation rate divided by the run length. Measure and record the power out of the 10% port of the optical fiber coupler, P_r(dBm). Connect the power meter in place of the receiver under test and adjust the attenuator so that the power meter again indicates P_r(dBm). The attenuator setting equals L_s(dB). Return the power meter to the coupler 10% output position.

30.4.5.1.2 Making a measurement. The input power to a DUT may be determined while performing any test method procedure without doing a direct measurement. Note the power meter indication, P_m(dBm), and the attenuator setting, A(dB). P_{in} (dBm), the average receiver input power level, may be calculated as:

$$P_{in}(dBm) = P_m(dBm)-[A(dB)-L_s(dB)]$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B

30.5 **OPTICAL POWER MEASUREMENTS**

30.5.1 Scope. This test method describes measurement of the average output power, extinction ratio and modulation rate capability of a digital fiber optic transmitter.

30.5.2 Symbols used.

DF_m: Duty factor of the envelope waveform of the transmitter output
DUT: Device under test
L_s: Reference established by the transmitter basic setup test method optical power measurement procedure
MR_H: Highest modulation rate required of the DUT
MR_L: Lowest modulation rate required of the DUT
P_m: Measured level of transmitter output power at the optical coupler 10% port
P_{out}: Transmitter output power level
V_{HR}: The top magnitude of the envelope waveform of a modulated optical carrier as defined by a referencing document
V_{LR}: The base magnitude of the envelope waveform of a modulated optical carrier as defined by a referencing document
XR: Extinction ratio

30.5.3 Test equipment.

30.5.3.1 Required. Test equipment required by this test method is that listed for the "power supply volatge and current" test method (see 30.2), the "basic setup for digital transmitter tests" (see 30.3) and in table IX. The capabilities required of test equipment associated with the referenced test methods may differ from those identified herein (see table IX).

Table IX. Required test equipment.

Item	Generic name	Capabilities	Suggested Mfr/Model
01	Digital oscilloscope		

30.5.4 Setup. The considerations and procedures of 30.1, 30.2 and 30.3 shall be observed or executed, as appropriate. The basic setup for digital transmitter tests (see 30.3) and digital oscilloscope are connected as shown in figure 7 to measure extinction ratio and average power. The basic setup and spectrum analyzer are connected as shown in figure 8 to measure modulation rate capability. The pattern shall be a square wave with duty factor 0.5 ± .05. Transition durations shall be not greater than 10% of the square wave period. Square wave frequency shall be variable over the entire modulation rate range identified by the referencing document.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the current version before use.

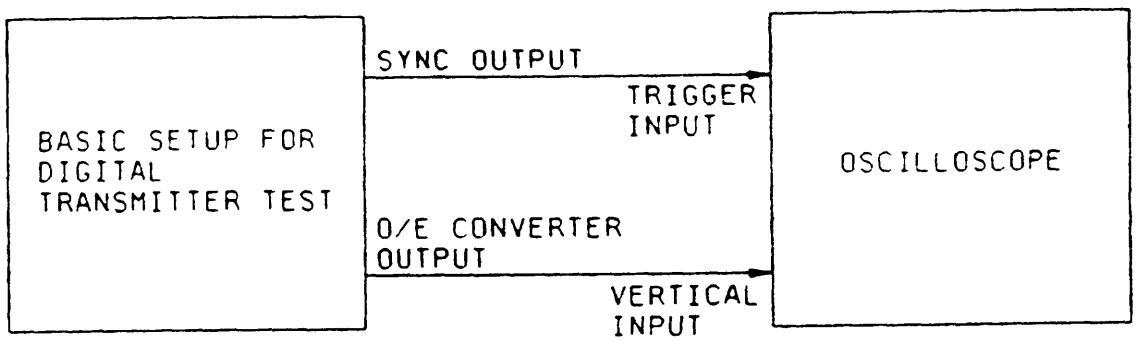


Figure 7. Output power and extinction ratio tests.

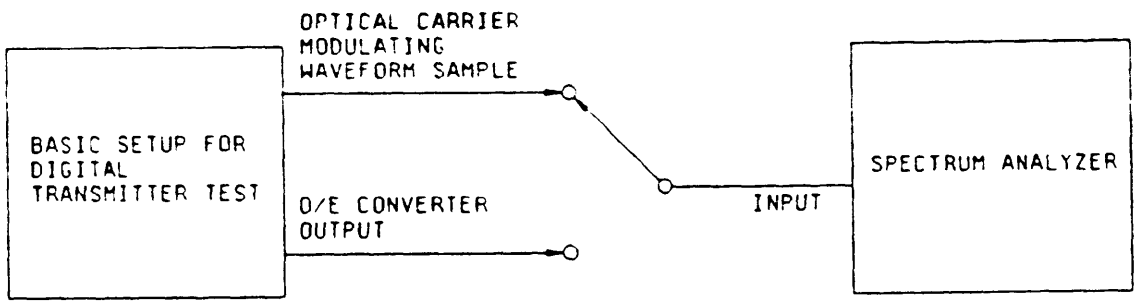


Figure 8. Modulation rate capability.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T02:42Z
Check the source to verify that this is the current version before use.

30.5.5 Procedures.

30.5.5.1 Output enable control. Some transmitters are provided with an input to disable the output. Average output power shall be measured with output enabled and disabled, where applicable. Modulation rate shall be measured with output enabled.

30.5.5.2 Extinction ratio.

30.5.5.2.1 Steps. Unless otherwise specified by the referencing document, set the square wave frequency to produce a modulation rate of $(MR_H+9*MR_L)/10$ where MR_L and MR_H are, respectively, the lowest and highest required modulation rates for the DUT. If run length is specified separately, MR_L shall be the lowest required modulation rate divided by the run length. Set oscilloscope vertical coupling to dc and display the waveform of the output from the optical to electrical (O/E) converter. Measure the waveform top (V_{HR}) and bottom (V_{LR}) magnitudes, defined by the referencing document, and duty factor, DF_m. Unless otherwise specified by the referencing document the reference level for determining duty factor shall be at 50% reference magnitude.

30.5.5.2.2 Analysis. Determine extinction ratio, XR, as:

$$XR = (V_{HR})/(V_{LR}).$$

30.5.5.3 Average power measurement.

30.5.5.3.1 Steps. Set the optical power meter (wavelength and level ranges) in accordance with the manufacturer's instructions to make an average power measurement. Note the measured value, $P_m(dBm)$.

30.5.5.3.2 Analysis. Calculate $P_{out}(dBm)$, the average transmitter output power, as follows:

$$P_{out}(dBm) = P_m(dBm) + L_s(dB) - 10*\log_{10}(2*[DF_m*(1-XR)+1]/[XR+1])$$

where $L_s(dB)$ is the reference established by the transmitter basic setup test method optical power measurement procedure. The last term corrects for duty factors that are not 0.50.

30.5.5.4 Verification of modulation rate capability.

30.5.5.4.1 Steps. Connect one of the optical carrier modulating waveform sample outputs from the transmitter test jig to the input to the spectrum analyzer. Insure that the analyzer is not being overloaded. Vary the modulation rate from the highest to the lowest required value by varying the square wave frequency. The variation should follow a sequence providing a resolution equal or better than that produced by a 1:2:5 ratio. Measure and record the level of the spectral component at the square wave frequency at each value. Connect the O/E converter output to the spectrum analyzer in place of the sample output. Repeat the procedure, varying the modulation rate in the same sequence. Measure and record the optical power meter indication as well as the level of the spectral component at the square wave frequency.

30.5.5.4.2 Analysis. Calculate the difference, in dB, between the levels measured at the modulating waveform sample output and the converter output for each of the modulation rates. Evaluation shall be in accordance with the requiring document. Unless otherwise specified by the requiring document, the upper limit on modulation rate for the DUT is that rate at which the value of the difference is 1 dB less than the value for 1/10 that rate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

30.6 **MODULATION ENVELOPE WAVEFORM MEASUREMENTS**

30.6.1 Scope. This test method describes measurement of the envelope waveform of the output from a digital fiber optic transmitter. The optical signal is converted to an electrical signal by an optical to electrical (O/E) converter and the measurements performed on the electrical waveform using a digital oscilloscope.

30.6.2 Symbols used.

DUT: Device under test
T_d: Pulse duration; applies to envelope waveform
T_p: Pulse period; applies to envelope waveform
UI: Unit interval; a time interval equal to the period of an associated clock
V_{HR}: The top magnitude of the envelope waveform as defined by the referencing document
V_{LR}: The base magnitude of the envelope waveform as defined by the referencing document

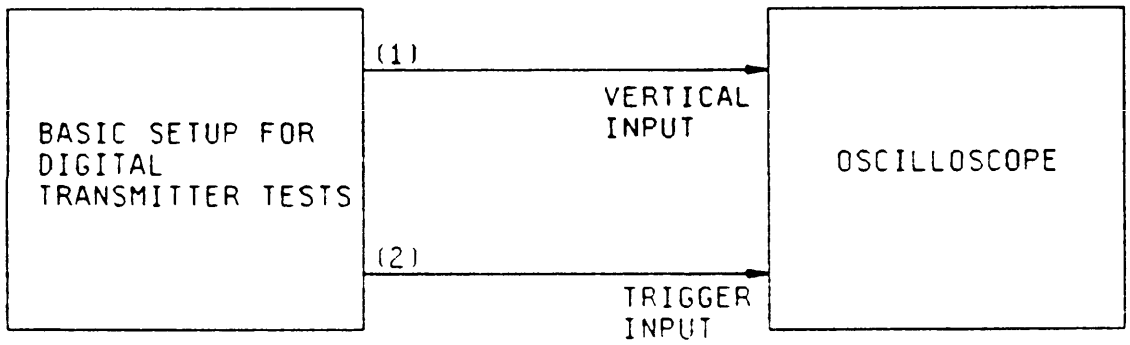
30.6.3 Test equipment.

30.6.3.1 Required. Test equipment required by this test method is that listed for the "power supply voltage and current" test method (see 30.2), the "basic setup for digital transmitter tests" (see 30.3) and in table X. The capabilities required of test equipment associated with the referenced test method may differ from those identified herein (see table X).

Table X. Required test equipment.

Item	Generic name	Capabilities	Suggested Mfr/Model
01	Digital oscilloscope		

30.6.4 Setup. The considerations and procedures of 30.1, 30.2 and 30.3 shall be observed or executed, as appropriate. Connect the basic setup for transmitter tests (see 30.3) and the oscilloscope as shown in figure 9. The requiring document must identify the portions of the waveform to be used to define transition waveform parameters and top and base magnitudes.



NOTES:
1. O/E converter output.
2. Pattern sync output.

Figure 9. Modulation envelope waveform measurement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

30.6.4.1 Conditions. Unless otherwise specified by the referencing document the settings and definitions of this paragraph apply. The pattern shall be a square wave. Modulation rate shall be $(MR_H+9*MR_L)/10$ where MR_L and MR_H are, respectively, the lowest and highest required modulation rates for the DUT. If run length is specified separately, MR_L shall be the lowest required modulation rate divided by the run length. Mesial magnitude is at 50% reference magnitude. The proximal line is at 10% reference magnitude and the distal line at 90% reference magnitude. The reference waveform is a trapezoidal waveform having transition durations and top and bottom magnitudes equal to those of the measured waveform.

30.6.5 Procedures.

30.6.5.1 Basic waveform measurements. Set clock frequency and pattern. Trigger the oscilloscope from the pattern generator sync output. Set vertical coupling to dc. Display the waveform of the optical to electrical (O/E) converter output signal on the oscilloscope. The portions used to define top and base magnitudes should include a minimum of twenty samples. Determine the waveform top (V_{HR}) and base (V_{LR}) magnitudes in accordance with the referencing document definition.

30.6.5.2 Transition durations. Display a transition waveform containing a base to top transition on the oscilloscope. Sample density should be sufficient to include not less than 10 samples between proximal and distal points. Use the top and base magnitudes determined above. Measure transition duration. Repeat the measurement for a top to base transition.

30.6.5.3 Bias distortion. Pulse duration (T_d) and pulse repetition period (T_p) shall be measured and noted. The difference, $|T_d-(T_p/2)|$, expressed as a percentage of a UI for the highest required modulation rate, is the (worse case) bias distortion. Evaluation of the DUT requires measuring the bias distortions of both the output from and input to the DUT. The bias distortion of the DUT is the difference between these two values. Since the input will usually be required to be compatible with a specified logic family, input bias distortion must be measured at a mesial level consistent with the reference level for that family.

30.6.5.3.1 Waveform distortions. Waveform distortions, such as preshoot, overshoot, tilt or ringing, are measured in terms of the magnitude of deviations from a reference waveform. Such deviations may be determined by subtracting the reference waveform from the waveform being measured. Deviation is expressed in percent, referred to the reference waveform.

30.6.5.4 Templates. The requiring document may express waveform requirements in terms of templates. In that case, conformance is determined through the use of those templates. Tolerances on bias distortion and transition durations interact so that separate templates are required to define top and base portions of the pulse waveform and transition waveforms, base to top and top to base.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

30.7 JITTER MEASUREMENTS

30.7.1 Scope. This test method describes procedures for measuring digital fiber optic transmitter and receiver jitter parameters. These parameters include jitter magnitude, internally generated jitter and jitter transfer.

30.7.1.1 Introduction. Measurement of jitter magnitude is fundamental to the determination of jitter characteristics. Two approaches, both employing general purpose test equipment, are described. Spectrum analyzer based procedures are described under "Device testing" (see 30.7.4.) in the body of the method and oscilloscope based procedures are described in section 30.7.5. The spectrum analyzer procedure is capable of measuring smaller values of jitter than the oscilloscope procedure but is limited to a maximum (peak to peak) amplitude approximately 10% of the test signal period. The generation of jitter values smaller than might be discernable with an oscilloscope is described in section 30.7.5.

30.7.1.1.1 Modulating waveform. Jitter is a continuous value, discrete time, function. This method treats the jitter waveform as a continuous value and time function.

30.7.1.1.2 Transition polarity. Jitter associated with positive (rising) and negative (falling) transitions of a waveform may not be equal. The procedures described herein do not explicitly address this. If the instrument used to measure jitter is sensitive to transition polarity (such as an oscilloscope) then data records should identify the polarity measured and, if the difference is significant, both transition polarities should be measured. The response of some instruments (such as a spectrum analyzer) does not distinguish polarity and indicates only a summation of the magnitudes associated with both polarities. Comparing measurements made with the two types of instruments, polarity sensitive and insensitive, requires summing the pair of values obtained with the former.

30.7.2 Symbols used.

d	difference between the level of a spectral line and the noise floor in a spectrum at a specific RBW (dB)
d(rms)	rms value of jitter (seconds) within a specified RBW
DUT	Device Under Test
ECW	Electrical Carrier Waveform (pattern generator output)
f_{clk}	clock generator frequency (Hz)
f_c	electrical carrier waveform frequency (Hz)
f_j	jitter modulation waveform frequency (Hz)
f(max)	frequency (Hz) at which J(max) occurs
JMW	Jitter Modulation Waveform (jitter modulator modulation input)
J(max)	maximum jitter amplitude (seconds, peak to peak) corresponding to LR(max)
J(ref)	sinusoidal amplitude (seconds, peak to peak) equivalent to the maximum random and systematic output jitter specified for DUT; (see 30.7.4.3)
j(rms)	value of rms jitter (seconds) for a specified bandwidth
LR(x)	level, in decibels, of the principal jitter spectral component relative to the level of the fundamental of the modulating waveform: $20\log_{10}(R(x))$ where x is not used, refers to the fundamental spectral component of the modulating waveform; x=f refers to the spectral amplitude at frequency f; x = ref refers to the value associated with J(ref); x = max refers to the maximum value attained during a measurement; x = 1 refers to a measured value
nep	noise equivalent power; optical power (watts) equivalent to the noise current at the input to a receiver
period	period of ECW (seconds)
PRBS	pseudorandom bit sequence
P(x)	optical input power (dBm) corresponding to V(x) (see 30.7.4.4) where, x = p refers characteristic plot parameter; x = r refers to reference condition; x = s refers to with stress applied;
P_o(avg)	average power (watts) of a modulated optical carrier input to or output from the DUT; the modulating waveform is assumed to have a constant value when averaged over some (short) period and multiples of that period

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the current version before use.

P_o(fund) that part of P_o(avg) which can be associated with the fundamental component of the modulating waveform
P_o(jitter) that part of P_o(avg) which can be associated with the jitter of the modulating waveform

R ratio of the level of a specified jitter spectral component to the level of the fundamental component of the modulation signal
RBW spectrum analyzer resolution bandwidth
R(x) spectrum analyzer reference level setting; x=1,2,...; (see 30.7.5.3.3.3.3)
V(x) rms voltmeter indication proportional to output jitter with P(x) optical input power; see 30.7.4.4
XR extinction ratio (dimensionless)

30.7.3 Test equipment.

30.7.3.1 Required. Test equipment required by this test method is that listed for the "power supply voltage and current" test method (see 30.2), the "basic setup for digital transmitter tests" (see 30.3), the "basic setup for digital receiver tests" (see 30.4) and in (table XI). The capabilities required of test equipment associated with the referenced test method may differ from those identified herein (see Table XI).

Table XI. Required test equipment.

ITEM	GENERIC NAME	CAPABILITIES	SUGGESTED MFR/MODEL
01	jitter modulation waveform source		
02	electrical attenuator		
03	oscilloscope		
04	adjustable delay		
05	spectrum analyzer		
06	signal generator		
07	optical receiver		
08	filter		
09	true rms voltmeter		
10	power supply signal injection circuits see 30.7.4		

30.7.3.2 Settings. Unless otherwise specified by the referencing document the following settings apply.

INSTRUMENT	PARAMETER	SYMBOL	SETTING
clock generator	frequency	f_{clk}	numerically equal to the highest modulation rate specified for the DUT
jitter modulator	modulation input	JMW	sine waveform
pattern generator	output pattern	ECW	square waveform
pattern generator	output frequency	f_c	equal to $f_{clk}/2$
pattern generator	output level		as required by jitter modulator to perform function
(na)	jitter frequency	f_j	range = from 10 Hz to, but not including, f_c

The waveform amplitude reference level for measuring jitter shall be the reference level for the associated waveform. The amplitude of the JMW is assumed to be independent of its frequency.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

30.7.3.3 Use.

30.7.3.3.1 General. The considerations and procedures of 30.1 shall be observed or executed, as appropriate. The DUT is to be mounted and powered as described by the basic setup method (see 30.3 or 30.4, as appropriate) and the power supply voltage and current test method (see 30.2). Instruments are not to be operated outside their linear range (overloaded). This applies especially to the spectrum analyzer and O/E converter or receiver. Take care during the procedures to avoid such operation.

30.7.3.3.2 Oscilloscope. Oscilloscope bandwidth should extend from dc to greater than three times f_c and sensitivity should be sufficient to produce full vertical deflection with the input signals. Controls shall be adjusted so that the waveforms to be evaluated are completely contained within and occupy the major portion (50% to 100% of full screen) of the display. The waveform should be centered in the display using time base, horizontal positioning and trigger delay (length of line between basic setup and trigger input) controls. Position the waveform so that the reference magnitude coincides with a major horizontal graticule line. Sweep rate should be sufficient to provide the horizontal resolution required for the measurements.

30.7.3.3.3 Spectrum analyzer. Unless otherwise specified, set spectrum analyzer resolution bandwidth (RBW) to not greater than 1% of the difference between the upper and lower limits of the specified frequency band. Relative levels are measured in decibels relative to the display reference level.

30.7.3.3.4 Filter. The purpose for the filter is to attenuate electrical carrier spectral components with respect to jitter spectral components. A multi-section LC low pass filter (LPF) having a cutoff frequency a few octaves below that of the square wave frequency is suggested. The addition of a notch filter, cascaded with the LPF, may be useful for reducing the fundamental component of the square wave.

30.7.3.4 Optical receiver. Receivers are usually more sensitive than laboratory o/e converters. Consequently, when evaluating transmitters the o/e converter of the basic setup is replaced by an analog optical receiver to measure the expected low level of jitter spectral density. The sensitivity and overload levels of the receiver must be evaluated to determine if they are adequate for this application.

30.7.3.4.1 Sensitivity. The spectral components of receiver noise must be negligible (-3 dB, preferably -10 dB) with respect to those due to the jitter to be measured. Assuming receiver noise spectral distribution to be uniform within the band to be measured, receiver capability may be evaluated on the basis of the value of nep, noise equivalent optical input power, for the receiver. Define an optical power, P_o(jitter), equivalent to that part of P_o(avg) associated with the jitter of the modulating waveform. P_o(avg) is the average power of the optical signal at the receiver input. The ratio of P_o(jitter) to P_o(fund), an optical power equivalent that part of P_o(avg) associated with the fundamental component of the modulating waveform, must not exceed the limit defined by:

$$\sqrt{2}\cdot\pi\cdot\frac{J(rms)}{(period)} = \frac{P_o(jitter)}{P_o(fund)} = \frac{nep}{P_o(fund)}$$

$$where\ P_o(fund) = \frac{\sqrt{2}}{\pi}\cdot\frac{(XR-1)}{(XR+1)}\cdot P_c(avg)$$

J(rms) is the maximum allowed jitter magnitude, period is the period of the square wave test signal, and XR is the transmitter extinction ratio. The criterion for assessing whether the measurement can be performed with a particular receiver is:

$$\left\{10\log_{10}\left[4\cdot\frac{J(rms)}{period}\cdot\frac{(XR-1)}{(XR+1)}\right] - 1.5\ dB\right\} > 10\log_{10}\left[\frac{nep}{P_c(avg)}\right]$$

A 3 dB (electrical) margin is allowed between the measured signal and receiver noise.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the current version before use.

30.7.3.4.2 Overload. The input level (dBm) at which a receiver becomes overloaded may be ascertained by evaluating the response at its output to a test signal applied to its input. The test signal is modulated by a square wave at frequency f_c. Use a spectrum analyzer to measure the difference, d, between the spectral amplitudes of the component at f_c and the noise within the receiver's bandwidth at the receiver output. The ratio of the change in d (dB, electrical) to the amount of a small change (a few dB, optical) in input level is evaluated. At very low input levels the ratio will be approximately two. As photodiode shot noise predominates over input amplifier thermal noise the ratio will approach one. Eventually a point is reached where further increase in input level results in the ratio becoming less than one. The receiver is said to be overloaded at that point. Since dynamic range is not increased above the level at which shot noise becomes dominant it is recommended that it not be exceeded.

30.7.3.5 Power supply signal injections circuits. These circuits are placed between the power supply and the power supply terminals of the device under test. They are used to impress the ripple/noise (stimulus) specified by the requiring document onto the power supply voltage and to provide a 50 ohm termination for the stimulus signal source. Schematics for these circuits appear in figure 16. The intent is for the power supply to appear as an equivalent series circuit consisting of a battery, the (equivalent) stimulus generator and an (approximate) one ohm series resistance in the frequency range specified by the requiring document. The supply side of R1 must be well bypassed so that negligible stimulus signal appears at that point.

30.7.4. Device testing. The procedures contained in this section are for testing devices (transmitters and receivers) to determine compliance with specified requirements.

30.7.4.1 Random and systematic jitter measurements. These procedures are based on the assumption that the jitter being measured has a Gaussian amplitude distribution and a wideband spectral distribution and that requirements are in terms of an rms limit value. Significant deviation from the distribution assumption will usually be evident in the spectral distribution. Spectrum analyzer RBW need not be less than 10 Hz. See 30.7.3.2. The JMW source is turned off for these procedures.

30.7.4.1.1 Spectral density. Spectral density is a measure of the concentration of jitter "power" (sec^2/Hz). The requiring document must identify a maximum limit and an associated frequency band. Measurements may employ limit levels, defined levels throughout the applicable frequency band which, if not exceeded, insure compliance with requirements.

30.7.4.1.1.1 Transmitters.

30.7.4.1.1.1.1 Setup. The basic setup for transmitter tests, described in method 30.3, and spectrum analyzer are connected as shown in figure 10. Replace the O/E converter of the basic setup with the optical receiver (see 30.7.3.4). The average optical input power to the receiver, P_o(avg), should be the maximum available which does not overload the receiver (see 30.7.3.4.2). Connect the receiver output to the spectrum analyzer input. Note that range extension (see 30.7.5.3.3.3) should not be used. A filter (see 30.7.3.2) may be introduced between the spectrum analyzer and the receiver output to reject those components not required for the measurement.

30.7.4.1.1.1.2 Procedure. The upper and lower limits of the frequency band to be examined shall be as defined by the referencing document. The frequency band may be broken into segments to facilitate measurement. Adjust the spectrum analyzer reference to place the peak of the spectral component at frequency f_c at the reference level. Measure and record the relative level, LR(f), of the spectrum at all frequencies in the frequency band being measured or, if limit levels are used, determine if they are exceeded. Remove the input to the spectrum analyzer and measure and record the level and distribution of the noise floor in the required band. The level of the jitter spectrum should be 3 dB greater than the analyzer noise floor throughout the frequency band. The measurement is limited by spectrum analyzer noise in those portions where this criterion does not apply.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the currentversion before use.

30.7.4.1.1.1.3 Analysis. Analysis is not required if limit levels are used. The value of the spectral density at a frequency is approximated by the ratio of the level of the jitter spectrum at that frequency to the resolution bandwidth used to measure that level. The value determined by the procedure (30.7.4.1.1.1.2) is relative to the square wave fundamental and needs to be converted to units of sec^2. Use the following formulas:

$$[d(rms)]^2 = \frac{1}{2}\left[\frac{R}{\pi}\cdot(period)\right]^2; \quad R = 10^{\frac{LR(f)}{20}}$$

$$SPECTRAL\ DENSITY = \sqrt{\frac{[d(rms)]^2}{RBW}}$$

where $[d(rms)]^2$ is the level of jitter "power" in units of sec^2 in the resolution bandwidth RBW and (period) is the period of the modulating waveform.

30.7.4.1.1.2 Receivers.

30.7.4.1.1.2.1 Setup. Connect the basic setup for receiver tests, described in method 30.4, and spectrum analyzer as shown in figure 10. Connect one of the outputs from the receiver data output interface circuits to the spectrum analyzer input. Insure that the interface output is properly terminated. See 30.7.3.2. Turn off the JMW source.

30.7.4.1.1.2.2 Procedure. Same as 30.7.4.1.1.1.2.

30.7.4.1.1.2.3 Analysis. Same as 30.7.4.1.1.1.3.

30.7.4.1.2 Rms jitter. The procedures used are the same as those used to determine spectral density. The purpose is to determine the total jitter "power" contained within a specified frequency band.

30.7.4.1.2.1 Setup. The setup used for spectral density measurements applies. See 30.7.4.1.1.1.1 or 30.7.4.1.1.2.1, as appropriate.

30.7.4.1.2.2 Procedure. The procedure used for spectral density measurements applies. See 30.7.4.1.1.1.2 or 30.7.4.1.1.2.2, as appropriate. The applicable frequency band shall be that identified by the referencing document. Determine the total power contained in the band or if a limit level is exceeded at any point (see 30.7.4.1.2.3) within the band.

30.7.4.1.2.3 Analysis. Compliance requires that rms jitter not exceed a limit specified by the requiring document. The value of rms jitter for the specified frequency band, j(rms), is computed by integrating the spectral density over that band:

$$j(rms) = \left[\int_{BW} \frac{[d(rms)]^2}{RBW}\right]^{\frac{1}{2}}$$

Alternately, limit level(s) may be established for all frequencies within the signal band. Compare the actual distribution to these limits to determine compliance.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

30.7.4.2 Pattern dependent jitter amplitude measurement. Jitter produced as a consequence of the pattern of the signal passing through a device is a function of that pattern and the response of the device to that pattern. Peak deviation is usually of primary interest. PRBSs or related patterns are often used for evaluation. Determination of peak displacement due to a PRBS can be difficult since the peak usually occurs only once within such sequences. Measurement can be facilitated by repeating the specific patterns that cause the peak deviations more frequently than they occur in a PRBS. These specific patterns need to be identified. Usually they represent the extremes of duty factor variation or intersymbol interference. The procedure described in this section is based on this approach. Results should be compared with those produced by conventional methods.

30.7.4.2.1 Setup. The setup shown in figure 11 is used. See 30.7.3.2.

30.7.4.2.2 Procedure. Perform this procedure twice, once for each of the patterns identified in 30.7.4.2.2.1 and 30.7.4.2.2.2. Turn off the JMW source. Set the pattern generator to produce the designated pattern repetitively. Set f_{clk} as required for the pattern. Set the oscilloscope to trigger from its external input and the trigger level to the reference level for the trigger input waveform. Set trigger slope to positive. Display the transition waveform (see 30.7.4.3.2). Measure and record the times at which the most early and most late traces intersect the reference horizontal graticule line with respect to that intersection by the bright center trace. Change the trigger slope to negative and repeat the measurement and recording of the trace intersection locations.

30.7.4.2.2.1 Intersymbol interference. This pattern is given in figure 12. The modulation rate shall be the maximum specified for the DUT.

30.7.4.2.2.2 Duty factor variation. This pattern is given in figure 13. The modulation rate shall be the minimum specified for the DUT.

30.7.4.2.3 Analysis. The bright center trace identifies the nominal location of the respective transition polarity (positive or negative) and the relative position of the other traces represents their individual displacements due to pattern dependent jitter.

30.7.4.3 Transfer measurements. This section describes procedures for measuring the ratio of jitter amplitude at a device output to the amplitude of the stimulus producing that jitter.

30.7.4.3.1 Input to output.

30.7.4.3.1.1 Setup. Connect the basic setup, spectrum analyzer, JMW source and electrical step attenuator as shown in figure 14. The amplitude of the test signal jitter modulation produced by the basic setup is assumed to be a function of JMW amplitude only. Unless otherwise specified by the referencing document the range applicable to f_j (transfer frequency band) shall be that for which the transfer parameter is specified. Unless otherwise specified by the referencing document the reference level, J(ref), for measurements of jitter amplitude shall be the peak to peak value of sinusoidal jitter equivalent to the maximum random and systematic jitter allowed for the DUT. Calculate the value of LR(jr), the log ratio of the spectral component at f_j to that at f_c for J(ref):

$$LR(jr) = 20\log_{10}\left[\frac{\pi}{2}\cdot\frac{J(ref)}{(period)}\right]$$

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the current version before use.

30.7.4.3.1.2 Procedure. Connect output (3) to the spectrum analyzer input. Adjust the spectrum analyzer to place the peak of the spectral component at f_c at the display reference level. Set f_j above its low limit by one tenth the difference between its low and high limits. Adjust the JMW level so that the level of the component at f_j is at LR(jr) with respect to the display reference. Measure and record the relative level and frequency of any other spectral components within the transfer frequency band which are greater than -10 dB relative to the component at f_j. Connect output (2) in place of output (3) to the spectrum analyzer input and readjust the spectrum analyzer reference to place the peak of the spectral component at f_c at the display reference level. Measure and record the relative level of the spectral component at f_j, LR(1). Return the output (3) to the spectrum analyzer input and reestablish the reference level. Vary f_j slowly from its low to its high limit to find the frequency at which the component at f_j achieves its maximum relative level. Measure and record this level, LR(max), and its associated frequency.

30.7.4.3.1.3 Analysis. Compute the difference (dB) between the maximum level at output (3) and the level at output (2), LR(max)-LR(1). This difference is the maximum input to output transfer magnitude in dB, which is given in terms of the transfer coefficient by:

$$\text{maximum transfer coefficient magnitude} = 10^{[\{L(max)-LR(1)\}/20]}$$

30.7.4.3.2 Power supply to output.

30.7.4.3.2.1 Setup. Connect the basic setup, spectrum analyzer, and signal generator as shown in figure 15. The test jig must incorporate circuits, such as those shown in figure 16, to provide for producing the power supply variations required by the specifying document. See 30.7.3.4. The power supply ac impedance is to be established determined by these circuits and should be approximately one ohm resistive. The signal generator waveform shall be sinusoidal. See 30.7.3.2. Turn off the JMW source. The data output from the DUT must be sensed differentially.

30.7.4.3.2.2 Procedure. Output level and frequency adjustments of the signal generator are assumed to be independent. Adjust the stimulus level to that which would produce the maximum voltage variation (peak to peak) specified by the referencing document at a power supply input terminal if the DUT and any allowed bypass capacitor were not present. DC voltage at the DUT power supply terminal shall be as specified by the referencing document. Adjust the spectrum analyzer reference to place the peak of the spectral component at f_c at the display reference level. Vary signal generator frequency slowly from the low to the high limit of the frequency band for which power supply transfer is specified while monitoring the amplitude of spectral components between 10 Hz and f_c. Measure the relative level, LR(max), and frequency of the spectral component attaining maximum amplitude.

30.7.4.3.2.3 Analysis. Convert the value measured for LR(max) to seconds (peak to peak) using the formula:

$$J(\text{max}) = \frac{2}{\pi} \cdot R \cdot (period); \quad R = 10^{\frac{LR(\text{max})}{20}}$$

Divide this value by the amplitude of the power supply variation and compare to the maximum value for power supply transfer allowed by the requiring document.

30.7.4.4 Monitor. It is often necessary to monitor the effect of a stress on a device. The parameter chosen to be monitored must be critical to the performance of the DUT and susceptible to the stress. The monitoring method needs to be uncomplicated. A convenient expression of the effect of the stress on a DUT is in terms of an equivalent change in input or output signal level. This usually satisfies the criticality criterion. The procedures are simplified as a consequence of the need to measure only the change between the unstressed and the stressed conditions, not the absolute values of the parameter used to monitor performance.

30.7.4.4.1 Transmitters. Assuming output power meets the susceptibility criterion, the effect of stress on a transmitter can be assessed by monitoring its change. A signal is applied to the DUT input to produce square wave modulation of the output at the maximum specified modulation rate.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the current version before use.

30.7.4.4.2 Receivers. Assuming that input signal level or internal gain meet the susceptibility criterion, the effect of stress on a receiver can be assessed by monitoring change in output rms jitter. Output jitter is particularly sensitive to input signal level below the sensitivity level. If the spectral distribution of the jitter is known to be invariant for the test conditions only a portion of the applicable frequency band need be monitored.

30.7.4.4.2.1 Setup. Connect the basic setup, filter and true rms voltmeter as shown in figure 17. See 30.7.3.2. Turn off the JMW source.

30.7.4.4.2.2 Procedure. Insure that all measurements are made in the linear range (see 30.7.4.4.2.2.2). There are two parts to this procedure, calibration of the DUT to determine the association between rms jitter and input signal level and the actual test.

30.7.4.4.2.2.1 Calibration. Measure and plot the voltmeter indication, V(p), produced at input signal level, P(p), for a sufficient number of points to produce a smooth curve. P(p) need only be measured as a relative value (attenuation). Do this for input powers between the sensitivity level and a level at which the voltmeter indication is within 3 dB of its response with no input to the DUT. The voltmeter indication is proportional to the amplitude of output signal jitter.

30.7.4.4.2.2.2 Test. Establish the DUT reference conditions. Adjust the input power to a level within the most sensitive portion of the voltage versus input power curve producing a voltmeter indication that is stable and substantially above any meter resolution or noise limit. Note the indication, V(r). Apply the stress to the DUT. Again note the voltmeter indication, V(s).

30.7.4.4.3 Analysis. Use the plot produced in 30.7.4.4.2.2.1 to determine the power levels P(r) and P(s) associated with the voltmeter indications V(r) and V(s) respectively. The change in input signal level is the difference P(r)-P(s).

30.7.5 Time domain measurements.

30.7.5.1 Time domain measurements. This section contains procedures for making time domain measurements of the amplitude of sinusoidal jitter. They may be used for comparison with other methods, to determine variation of amplitude and phase with frequency, applied to more complex waveforms and to calibrate instruments for measuring or generating jitter.

30.7.5.2 Setup. Required equipment and their interconnection are indicated in figure 11.

30.7.5.3 Procedures.

30.7.5.3.1 Basic. Set f_{clk}, f_c, f_j and pattern as specified (see 30.7.3.2). Set JMW source level to minimum and waveform to sinusoidal. Set the step attenuator to minimum attenuation. Connect the (unjittered) reference sample signal to the oscilloscope external trigger input and the signal to be evaluated to a vertical input. Set the oscilloscope to trigger from its external input and the trigger level to the reference level for the trigger input waveform. Set trigger slope to display the selected transition polarity. Triggering the oscilloscope from its external input establishes the unmodulated signal as reference and is in accordance with the definition for jitter which establishes "the nominal transition position" as reference. The procedures discussed in this section are based on the use of a conventional analog oscilloscope. Other instruments might be used with appropriate modifications to the procedures. Display a transition waveform (see 30.7.3.3.2). Adjust the JMW source level so that the width of the intersection of the waveform envelope by the horizontal reference line is approximately 35% of the clock period.

30.7.5.3.2 Linearity.

30.7.5.3.2.1 Purpose. The linearity of the transfer between stimulus input (jitter modulation) to a device and the resulting jitter at its output is determined.

Source: https://assist.dla.mil -- Downloaded: 202640-20T02:42Z
Check the source to verify that this is the current version before use.

30.7.5.3.2.2 Steps. Perform the basic procedure (30.7.5.3.1). Measure the horizontal extent of the intersection of the waveform envelope with the reference horizontal graticule line (increase sweep rate, if necessary). Record the extent and associated step attenuator setting. Increase the step attenuator setting by about 2 or 3 dB. Repeat this measure-record-increase sequence until there is no significant change in the measured extent.

30.7.5.3.2.3 Analysis. Calculate the ratio, for each step attenuator setting, of the extent to that for the preceding setting. Express this ratio in dB ($20*\log_{10}(\text{ratio})$). Compare the calculated value to the change in attenuation (current step attenuator setting - previous step attenuator setting). The difference between the ratio and the change in attenuation is a measure of the deviation from linearity plus measurement errors. See 30.7.5.3.3.2 ("random waveform").

30.7.5.3.3 Magnitude measurement.

30.7.5.3.3.1 Coherent waveform.

30.7.5.3.3.1.1 Purpose. The peak to peak amplitude of sinusoidal jitter is determined.

30.7.5.3.3.1.2 Steps. The horizontal extent measured in the linearity procedure (30.7.5.3.2) is the peak to peak amplitude of the sinusoidal jitter. That procedure can be used to determine the peak to peak amplitude of most large amplitude coherent jitter. If the linearity of a process is established by the procedure above, the process may be used to provide a signal with a known small amount of jitter by reducing the amplitude of the modulating signal by a known amount of attenuation. An unknown jitter amplitude may be determined by comparing it with a known amplitude. See 30.7.5.3.3.2 ("random waveform").

30.7.5.3.3.2 Random waveform.

30.7.5.3.3.2.1 Purpose. The rms value of random jitter, such as that appearing at a receiver output, is determined. Random jitter is difficult to measure with an oscilloscope because the boundaries of the trace envelope are not distinct. This procedure helps to reduce the subjectivity of such measurements.

30.7.5.3.3.2.2 Steps.

30.7.5.3.3.2.2.1 Preparation. Set the JMW to a square wave and its frequency, f_j, low with respect to f_c but sufficiently above any low frequency limit to avoid significant distortion. Calibrate the jitter produced by the JMW as described in 30.7.5.3.3.1 ("coherent waveform").

30.7.5.3.3.2.2.2 Measurement. Set the step attenuator to its maximum value. Decrease the step attenuator setting until a display consisting of two envelopes separated by an approximately equal width dark area is obtained. Continue to decrease the attenuation just to the point at which the dark area is no longer discernible. Record the peak to peak amplitude of the impressed jitter.

30.7.5.3.3.2.3 Analysis. The peak to peak amplitude of the impressed jitter is equal to twice the magnitude of the residual jitter. The meaning of magnitude depends upon the residual jitter waveform. If the residual jitter has a gaussian amplitude distribution then the magnitude is approximately its rms value. Linearity measurements (30.7.5.3.2) can be corrected for such residual jitter by subtracting (rms subtraction) its value from the amplitudes measured in the linearity procedure.

30.7.5.3.3.3 Range extension. This method is presented in terms of the spectrum analyzer procedure described in "device measurements" (see 30.7.4.). It may be used with other methods to isolate jitter components for measurement and analysis.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

30.7.5.3.3.3.1 Discussion. The range over which jitter magnitude can be measured is limited by the dynamic range of the measuring instrument. This range can be extended to lower values by subtracting the undistorted electrical carrier waveform (test signal) from the waveform at the output of the device under test. The method is illustrated by figure 18. The most critical function is wave shaping. Waveforms identical in amplitude and transition waveform and with transition locations dependent solely on the associated input need to be generated from the two inputs. The test signal is divided and passed through two paths, one through the test setup and the other through a delay equal to the average delay through the test setup. The threshold adjustments of the wave shaping function are used to equalize the average regenerated pulse widths. The output from the SUM function is the summation of its two input waveforms. The fundamental component is suppressed and the average value of the output is proportional to the test setup output jitter. The SUM output also contains undesired components due to unintended amplitude and time differences between the regenerated waveforms and imperfections in the balance of the SUM circuit. The TEST output is equal to the output from the test setup with respect to jitter. The fundamental level may be measured at the TEST output and the level of the jitter components at the SUM output. Thus the dynamic range limitation that exists when the test setup output is measured directly is reduced.

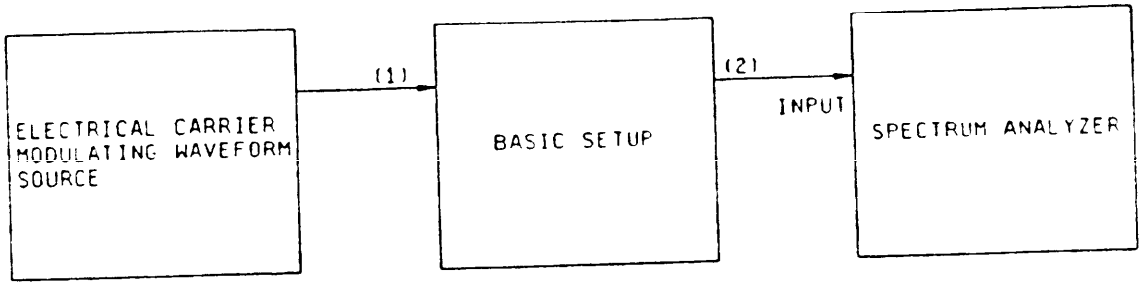
30.7.5.3.3.3.2 Setup. Use the setup shown in figure 19.

30.7.5.3.3.3.3 Steps. Set clock, JMW source, attenuator and pattern generator as directed in 30.7.5.1.2.1. Turn off the JMW source. Set the SUM threshold adjustments to the reference level for the respective input waveforms. Monitor the SUM output with the spectrum analyzer and adjust the delay and thresholds to minimize the level of the spectral component at f_c in the SUM output. Monitor the TEST output with the spectrum analyzer. Set the peak of the spectral component at f_c to the display reference level. Turn on the JMW source and set the peak of the spectral component at f_j to a level which is +20 dB to +30 dB with respect to the adjacent noise. Record this peak level with respect to the display reference, L(p). Monitor the SUM output with the spectrum analyzer and, using the analyzer reference level adjustment, set the peak of the spectral component at f_j to the display reference level. Turn off the JMW and determine the level, L(m), of the jitter components with respect to the display reference (see 30.7.4.1.1 and 30.7.4.1.2).

30.7.5.3.3.3.4 Analysis. Determine the level of the jitter components relative to the peak of the f_c spectral component by calculating L(m)+L(p).

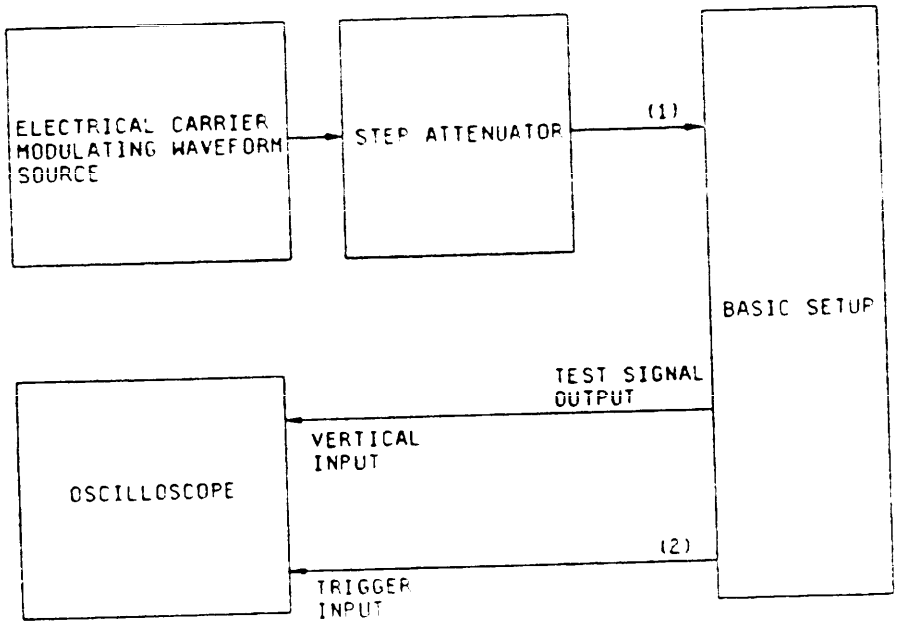
Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B



NOTES:
1. Modulation input to jitter modulator.
2. Transmitter tests: O/E converter (receiver) output.
   Receiver tests: data output from the receiver test jig.

FIGURE 10. Jitter special density and amplitude measurements.

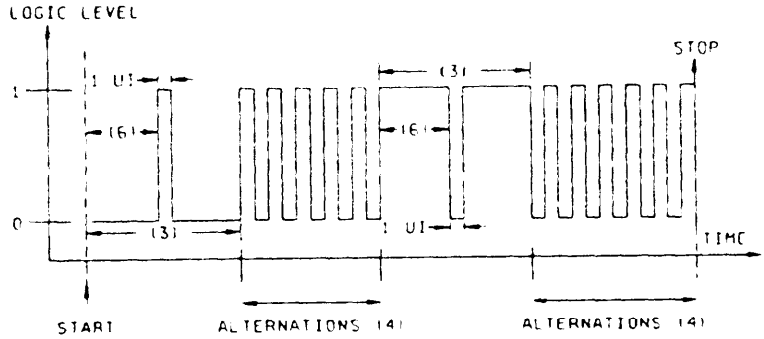


NOTES:
1. Modulation input to jitter modulator.
2. Pattern dependent jitter measurements: optical carrier modulating waveform sample output.
   Time domain jitter measurement: pattern generator sync output.

FIGURE 11. Pattern dependent and time domain jitter measurements.

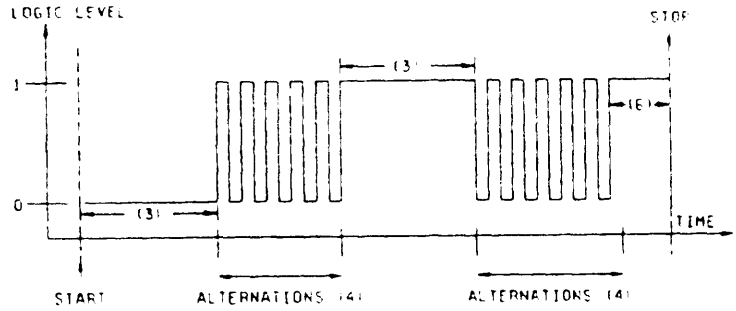
Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B



NOTES:
1. Modulation rate is maximum.
2. UI = unit interval.
3. Maximum run length plus two unit intervals.
4. Alternations refer to pattern alternating between logic level "1" and logic level "0" in succeeding unit intervals. The combined length of both alternation sequences equals two times the maximum run length less two unit intervals.
5. The complete pattern is repeated continuously, with the stop times of one pattern being the start time of the succeeding pattern.
6. The nearest even number of unit intervals equal to or greater than one half the maximum run length.

FIGURE 12. Intersymbol interference test pattern.

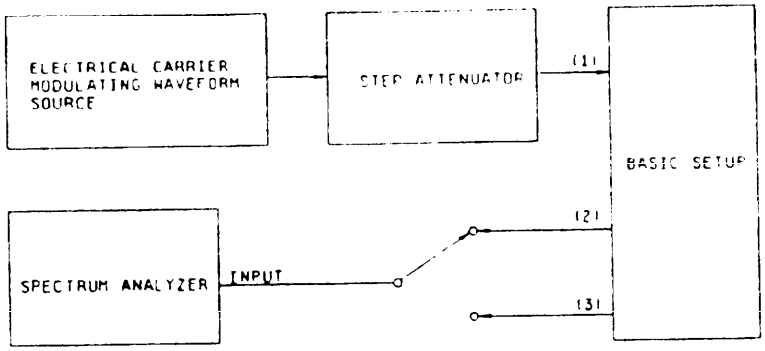


NOTES:
1. Modulation rate is minimum.
2. UI = unit interval.
3. Maximum run length.
4. Alternations refer to a pattern alternating between logic level "1" and logic level "0" in succeeding unit intervals. The combined length of both alternation sequences is two times the maximum run length less one unit interval.
5. The complete pattern is repeated continuously, with the stop time of one pattern being the start time of the succeeding pattern.
6. The length of this interval is used to adjust the duty factor of the complete pattern to be as close as possible to the maximum value. The pattern is inverted to produce the minimum duty factor.

FIGURE 13. Duty factor test pattern.

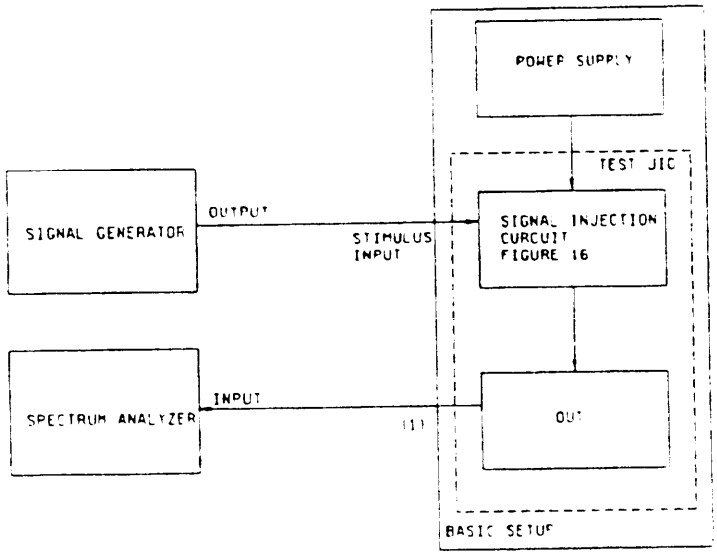
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B



NOTES:
1. Modulation input to jitter modulator.
2. Transmitter tests: optical carrier modulating waveform sample output.
   Receiver tests: O/E converter output.
3. Transmitter tests: O/E converter (receiver) output.
   Receiver tests: data output from receiver test jig.

FIGURE 14. Jitter transfer, input to output.

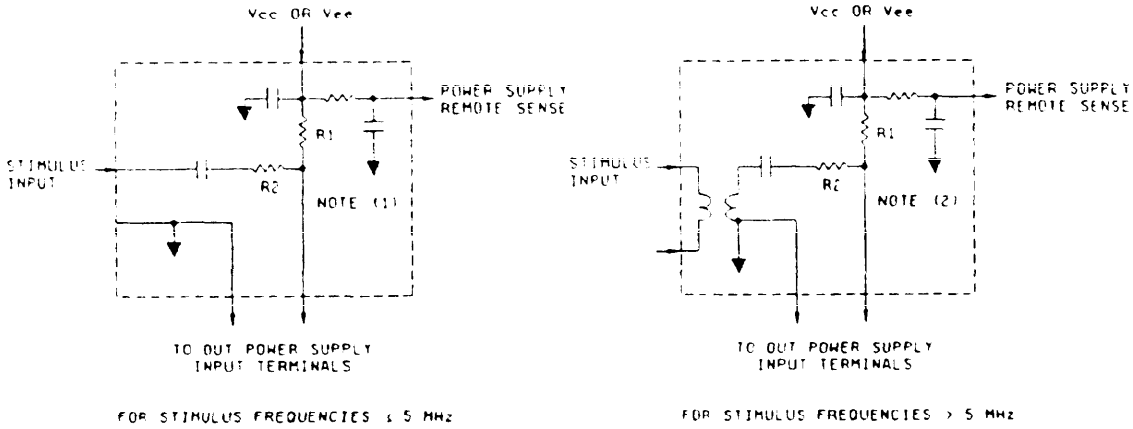


NOTE:
1. Transmitter tests: O/E converter (receiver) output.
   Receiver tests: data output from receiver test jig.

FIGURE 15. Power supply noise rejection measurement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B



NOTES:
1. R1 = 1Ω; R2 = 50Ω.
2. R1 = 1Ω; R2 = 11.5Ω.
3. 4:1 impedance ratio.

FIGURE 16. Power supply noise rejection signal injection circuits.

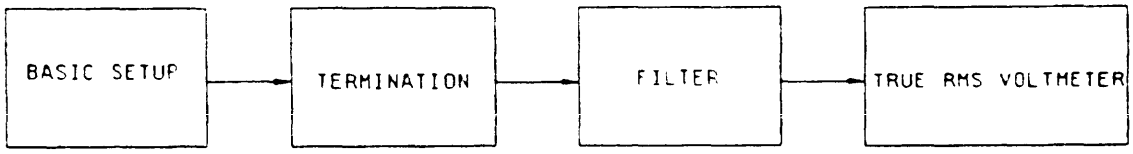
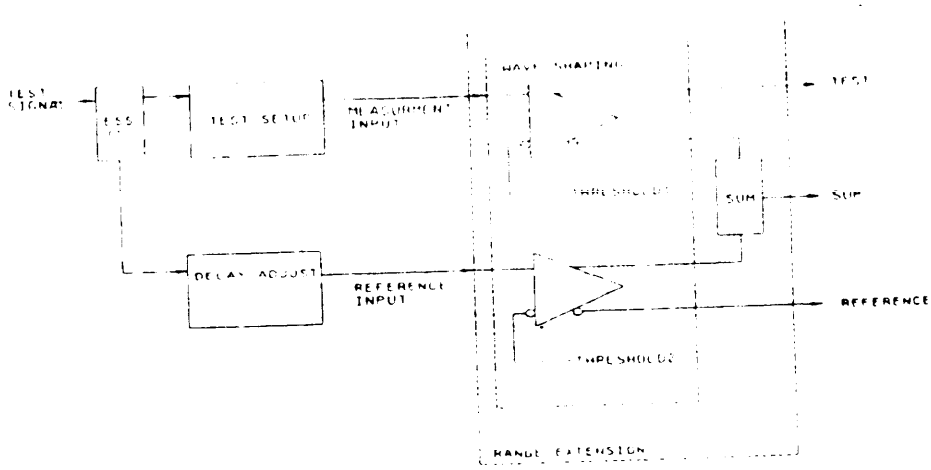


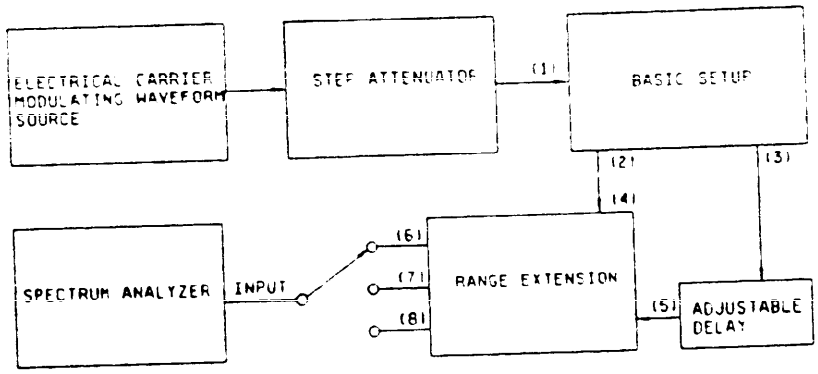
FIGURE 17. Jitter monitor measurements.

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T02:42Z
Check the source to verify that this is the currentversion before use.



NOTE:
1. ESS = electrical signal splitter.

FIGURE 18. Range extension unit.



NOTES:
1. Modulation input to jitter modulator.
2. Transmitter tests: O/E converter output.
   Receiver tests: data output from receiver test jig.
3. Sample output from jitter modulator.
4. Measurement input.
5. Reference input.
6. Test output.
7. Sum output.
8. Reference output.

FIGURE 19. Range extension measurement.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B

30.8 DIGITAL RECEIVER BER MEASUREMENT

30.8.1 Scope. This test method describes measurement of bit error ratio (BER) for digital fiber optic receivers. A pattern generator and error detector are used.

30.8.2 Symbols used.

B: Amount of pattern bias (a uniform deviation of pattern transitions of the same polarity from their nominal locations; see FED-STD-1037A) expressed herein in terms of time
BER: Bit error ratio
DT: Delay time; the set value for a variable delay element
DT1: Set value for a variable delay elements determining one of the limits on a range of delay times producing an unstable output from an error detector input sampler
DT2: Same as DT1
LPF: Low pass filter
UI: Unit interval

30.8.3 Test equipment.

30.8.3.1 Required. Test equipment required by this test method is that listed for the "power supply voltage and current" test method (see 30.2), the "basic setup for digital receiver tests" test method (see 30.4) and in table XII. The capabilities required of test equipment associated with the referenced test methods may differ from those identified herein (see table XII and 30.8.3.2)

Table XII. Required test equipment.

Item	Generic name	Capabilities	Suggested Mfr/ Model
01	Error detector	see 30.8.3.2 herein	
02	VARIABLE DELAY 1	see 30.8.3.3 herein	
03	VARIABLE DELAY 2	see 30.8.3.3 herein	
04	VARIABLE DELAY 3	see 30.8.3.3 herein	
05	Oscilloscope		
06	Resistive pad	50 ohm impedance; 6 dB loss	
07	Low pass filter	50 ohm impedance	
08	True rms voltmeter		

30.8.3.2 Error detector. The functions to be provided by the error detector are illustrated by the block diagram shown in figure 20. The interface circuits at the inputs convert between the incoming signal levels and the logic levels internal to the error detector. The buffer amplifiers at the outputs provide for driving a 50 ohm load. The two input samplers provide for independent sampling of the two input channels. The reference and measurement input patterns are compared by the exclusive OR to detect errors. The output from the exclusive OR is sampled so that only one pulse of width one unit interval is produced for each error. The error detector recovers neither the clock nor the pattern. Consequently, testing can not be interrupted by "sync loss" and high rates of error can be measured.

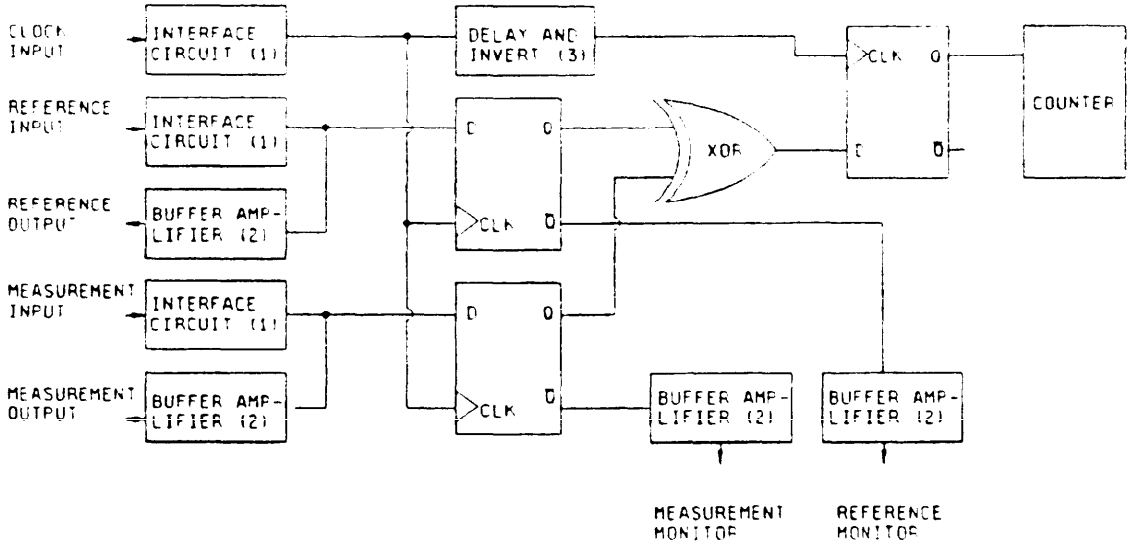
30.8.3.3 Variable delays. Three variable delay elements are used to align and sample the reference and measurement signal patterns. The delay of VARIABLE DELAY1 should be slightly greater than that between the pattern generator output and the data output from the receiver under test. It is best made up of a series combination of a fixed and a variable delay to provide the required setting resolution and stability. VARIABLE DELAY2 and VARIABLE DELAY3 should have a range of approximately 2 unit intervals and resolution of about 0.05 unit interval.

30.8.3.4 Stability indicator. The procedure described in 30.8.5.2, herein, requires that the center of a range of delay settings producing an unstable output from an error detector input sampler be determined. An indication of the amount of instability may be obtained by monitoring the noise level in a relatively narrow frequency band at the sampler output. A wideband true rms voltmeter is used to measure noise amplitude and a low pass filter (LPF) to define the measurement bandwidth. This is shown in figure 21. A pad is used to provide isolation between the filter and the output. The LPF should have a cutoff frequency approximately one tenth that of the frequency of the square wave test pattern.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the currentversion before use.

30.8.5.3.1 Pattern sensitivity determination. Determine the average input power (dBm) required to produce a BER three orders of magnitude greater than that associated with the required sensitivity. Use the square wave pattern. Record the power level and resultant BER. Repeat this measurement using three patterns. The intersymbol interference pattern and duty factor pattern and its inversion (see 3.7.4.2.2 and 3.7.4.2.2.2.). The modulation rate for the intersymbol interference and square wave patterns shall be the maximum specified for the DUT. The modulation rate for the duty factor patterns shall be the minimum specified for the DUT. The input signal must provide the same modulation amplitude as used for the square wave pattern measurement. Therefore, switch to the square wave pattern when setting or measuring input power. Compare the measured BER values for each of the four patterns for each device and determine which pattern produces the greater value. Determine and note the input power required to produce this greater value of BER with the square wave pattern. Calculate the relative difference (dB) between this input power and that used for the initial measurement. This difference is the pattern sensitivity of the DUT and shall not exceed the maximum allowed by the requiring document.

30.8.5.3.2 Measurement. Measure and note BER for each device. Use the pattern identified in 30.8.5.3.1 as producing the greater value of BER. Set input power to that associated with the specified sensitivity for the DUT. Measure input power with the square wave pattern. The measured BER shall not exceed maximum allowed for the DUT.



NOTES:
1. Interface circuits convert input signal levels to logic levels required by error detector circuits.
2. Buffer amplifiers have a gain of +1 and provide capability to drive a 50 Ω load.
3. As required so that exclusive OR output is clocked to counter at a time midway between the times that input data is clocked into the error detector.

FIGURE 20. Error detector functional diagram.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

30.8.4 Setup. The considerations and procedures of 30.1 and 30.2 shall be observed or executed, as appropriate.

30.8.4.1 General. The setup is illustrated in figure 21. The receiver basic setup (see 30.4) is used. The error detector (see 30.8.3.2) compares the received to the reference data identify errors. The pattern waveform sample output is used for the reference. Connections between the error detector and the oscilloscope should be made with cables having equal time delays. Oscilloscope internal delay between vertical channels should be reduced to a negligible amount.

30.8.4.2 Settings. Unless otherwise specified herein, set the pattern generator to produce a square wave at the maximum modulation rate specified for the DUT. Trigger the oscilloscope from the pattern generator sync output. Unless specified herein, error detector inputs are to be sampled at the center of the bit interval.

30.8.5 Procedures. Unless otherwise specified below, set the receiver input level so that BER is negligible (less than 1E-6). Set VARIABLE DELAY2 and VARIABLE DELAY3 to mid range.

30.8.5.1 Reference pattern delay adjustment. The propagation times from the output of the pattern generator to the reference and measurement inputs to the error detector must be made equal. The reference signal is delayed to make up the difference between the two paths. Set the pattern generator to produce a repeated pattern having a duty factor of 0.5 and a unique and a readily identifiable feature at the maximum specified modulation rate for the DUT such as the pattern identified in 30.10.5.1. The pattern length should be greater than the maximum expected difference between the propagation times with 0 delay to avoid ambiguity. Temporarily connect the error detector reference and measurement input signals to individual vertical inputs to the oscilloscope and display their waveforms. Display the unique pattern feature of the measurement waveform. Adjust VARIABLE DELAY1 to align the unique feature of the reference waveform with that of the measurement waveform in the oscilloscope display at the reference amplitude for the error detector.

30.8.5.2 Sampling instant.

30.8.5.2.1 Location. This procedure is used to determine the locations of transitions at the inputs to the error detector reference and measurement signal samplers with respect to active edges at its clock input. Use the square wave pattern. Connect the monitor output from the sampler to be calibrated to the input to the stability indicator (see 30.8.3.4). Slowly increase VARIABLE DELAYλ (λ =2 or 3), as appropriate from a setting of 0 to determine the first, second, third and fourth settings at which the voltmeter indicates a peak. Note these settings, DT1, DT2, DT3, and DT4. These correspond to transition time locations and define three adjacent unit intervals. The differences DT3-DT1 and DT4-DT2 should equal twice a unit interval.

30.8.5.2.2 Bias correction. Bias is an expression of the uniform displacement of (either rising or falling) transitions relative to the complementary transition type. The magnitude of the bias, B, is $|[(DT3-DT1)/2]-(DT2-DT1)|$. Bias is to be equally distributed at the nominal transition locations by displacing the ideal locations by B/2. For example, to sample the center of the first interval, delay shall be set at (DT1+DT2)/2. The location for the succeeding interval is (DT2+DT3)/2.

30.8.5.3 Bit error ratio (BER). BER is defined as

BER = (# ERRORS ACCUMULATED DURING TEST TIME)/(TEST TIME)

Tests shall be run until 100 errors are accumulated or for a time equal to 100/(BER*DATA RATE) where value of BER is that associated with the sensitivity specified for the DUT for the condition being evaluated.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

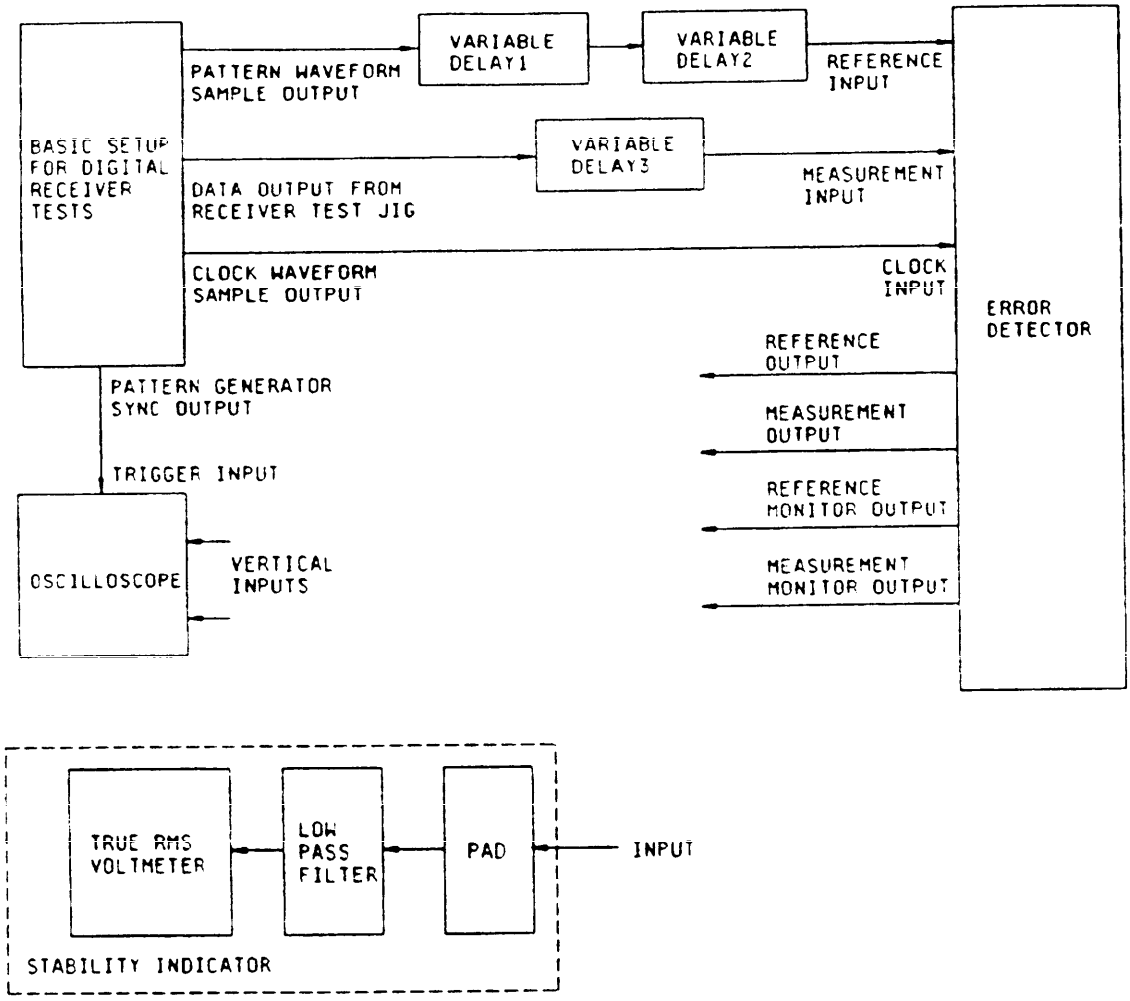


FIGURE 21. Receiver bit error ratio measurements.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B

30.9 CORRELATION BETWEEN BER AND JITTER MEASUREMENTS

30.9.1 Scope. This test method describes two procedures to evaluate signal quality at the output of a digital fiber optic receiver. Results are comparable to BER measurements but can be obtained in significantly less time. One of the procedures is based on the actual relationship between jitter and BER for the receiver design being evaluated and the other on a theoretical relationship between those parameters.

30.9.2 Symbols used.

BER: Bit error ratio
P(sw): The input power level associated with the sensitivity specification for the DUT reduced by the pattern sensitivity.
dj: Change in output jitter between an input level producing a BER of 10^{-6} and P(sw) for a square wave pattern.

30.9.3 Required test equipment. Refer to 30.8, the digital receiver BER measurement test method.

30.9.4 Setup considerations. The setup for the receiver BER measurement test method, 30.8, shall be used.

30.9.5 Procedures. The decrease (dB) in output jitter produced by increasing optical input power from the level associated with a BER of 10^{-6} to that associated with the rated sensitivity is determined. This decrease is compared to the increase in s/n required to reduce BER from 10^{-6} to the required value.

30.9.5.1 Determining the relative decrease in jitter amplitude.

30.9.5.1.1 Direct measurement. Measure output BER and the jitter of the DUT at each input level in an unbroken stepped sequence of input levels. Steps shall be 0.25 dB and shall start from a level equal to or less than that yielding a BER of 10^{-6} and proceed to a level equal to or less than P(sw). P(sw) is the input power level associated with the sensitivity specified by the referencing document. Use the bit error ratio measurement procedure of the digital receiver BER measurement test method, 30.8, with a square wave pattern. See the jitter measurement test method, 30.7, for jitter measurement procedures. Record the input levels (dBm) and associated BERs and jitter levels. Use this data to determine dj, the change (dB) in jitter between input levels producing BERs of 10^{-6} and P(sw), for the DUT. Interpolation between measurement data points is acceptable.

30.9.5.1.2 Inferred measurement. This procedure is based on the theoretical relationship between error and signal to noise ratios for binary signals with added random noise. It may be used in place of 30.9.5.1.1, herein, if permitted by the requiring document. Determine the theoretical ratio (in dB), I, of the (electrical) s/n required to produce the BER required by the referencing document to that required to produce a BER of 10^{-6}
(see 30.9.3.4).

30.9.5.2 Evaluation. This procedure rapidly evaluates output signal quality on devices. Actual performance, as determined by performing the direct measurement procedure, 30.9.5.1.1, or theoretical performance as determined in accordance with 30.9.5.1.2 (and accepting a penalty), is used as a criterion for all devices. The individual device evaluation error ratio measurement procedure of the receiver BER measurement test method, 30.8, is used. Set the input level to produce a BER of 10^{-6}. Measure and record the value of the output jitter using the same method as used in 30.9.5.1.1. Increase input level to reduce output jitter by dj, as identified by referencing document, or (I+1) dB, if the inferred method is allowed. Input power shall be not greater than P(sw).

30.9.5.3 Reference. A discussion of theory describing a binary signal in the presence of random noise and curves and tables relating s/n and BER can be found in many texts. One such text is "Digital Transmission Systems" by P. Bylanski and D.G.W. Ingram published by Peter Peregrinus Ltd. in 1976, chapter 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B

30.10 **OPTOELECTRONIC DELAY MEASUREMENT AND PHASING DETERMINATION**

30.10.1 Scope. This method describes a procedure to measure propagation delays and phasing between inputs and outputs of digital fiber optic transmitters and receivers.

30.10.2 Symbols used. The "x" in the subscripts of the listed symbols is replaced by "r" or "f", as appropriate to indicate that the parameter is associated with a rising or falling input transition, respectively.

D_{eox}	propagation delay from input to output for the E/O converter
D_{oex}	propagation delay from input to output for the O/E converter
D_{rx}	propagation delay from input to output for a fiber optic receiver module
D_{ss}	propagation from input to output of the electrical signal splitter
D_{tx}	propagation from input to output for a fiber optic transmitter module
DUT	device under test
VD_{sx}	value of variable delay setting measured for the setup measured with either the E/O or O/E converter in place of the DUT test jig (see 30.10.5.3)
VD_{rx}	value of variable delay setting measured for the setup with the receiver test jig in place (see 30.10.5.4.2)
VD_{tx}	value of variable delay setting measured for the setup with the transmitter test jig in place (see 30.10.5.4.2)

30.10.3 Test equipment.

30.10.3.1 Required. Test equipment required by this test method is that listed for the "power supply voltage and current" test method (see 30.2), the "basic setup for digital transmitter tests" (see 30.3), the "basic setup for digital receiver tests" (see 30.4) and in table XIII. The capabilities required of test equipment associated with the referenced test method may differ from those identified herein (see table XIII).

Table XIII. Required test equipment.

ITEM	GENERIC NAME	CAPABILITIES	SUGGESTED MFR/MODEL
01	clock generator	see 30.3	
02	pattern generator	see 30.3	
03	transmitter test jig	see 30.3	
04	variable delay		
05	electrical signal splitter	see 30.3	
06	E/O converter	non-inverting, see 30.3	
07	optical attenuator	see 30.3	
08	cladding mode stripper	see 30.3	
09	O/E converter	non-inverting, see 30.3	
10	receiver test jig	see 30.4	
11	oscilloscope		

30.10.3.2 Converter delays. The values of the propagation delays for the E/O and O/E converters, D_{eox} and D_{oex}, respectively, must be known. If they are not known, half the value of the total delay for a series connection of the E/O and O/E converters may be assigned to each. The propagation delay through the electrical signal splitter, D_{ss}, must be known and the difference between the delays to each of the outputs should be negligible.

30.10.4 Setup considerations. The considerations and procedures of 30.1 and 30.2 shall be observed or executed, as appropriate. The setups for transmitter and receiver tests are as shown in figures 22 and 23, respectively.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

the applicable associated detail specification(s). New quality assurance programs which are initiated by

30.10.5 Procedures.

30.10.5.1 Preliminary settings. Set the pattern generator to produce a repeated waveform in compliance with the pattern requirements for the device under test and having a duty factor of 0.5, a unique and readily identifiable feature and a period greater than two times the total time required for the pattern generator signal to pass through the DUT and arrive at the oscilloscope input. An example of such a pattern is shown in figure 24.

30.10.5.2 References. Times are measured between an input transition and a corresponding output transition. The corresponding transition may be of the same or opposing polarity (rising or falling). Unless otherwise specified by the referencing document, the magnitude reference for logic waveforms shall be the reference for the associated logic family and that for optical waveforms shall be at the associated mesial level.

30.10.5.3 Determining setup phasing and delay. Connect the electrical signal splitter and E/O converter combination or the O/E converter, as appropriate, in place of the DUT test jig. See the applicable figure, 22 or 23. Display the waveform of the signal received from the o/e converter or receiver under test, as appropriate, on the oscilloscope. Display the waveform of the signal received from the electrical signal splitter. Verify that the phasing of the two waveform patterns is non-inverting. Adjust the variable delay to align the start/stop lines corresponding to the leading edge of the unique feature in each waveform. Record the variable delay setting, VD_{sx}. Repeat for the trailing edge.

30.10.5.4 Determining transmitter and receiver phasing and delay.

30.10.5.4.1 Reconfiguration and phasing. Restore the DUT test jig in place of the electrical signal splitter and E/O converter combination or the O/E converter, as appropriate. Make the appropriate electrical connection to the transmitter input or receiver output being evaluated. Reestablish the oscilloscope displays as in 30.10.5.3. Compare the two waveforms to determine DUT phasing (inverting or non-inverting). Record the phasing.

30.10.5.4.2 Delay measurement. Readjust the variable delay, as described in 30.10.5.3, to align the stop/start lines corresponding to the transitions being measured. Record the variable delay setting (VD_{tx} for transmitter and VD_{rx} for receiver measurements).

30.10.5.4.3 Analysis. The transmitter and receiver optoelectronic delays associated with rising and falling input transitions can be calculated from the variable delay setting values determined in 30.10.5.4.2.

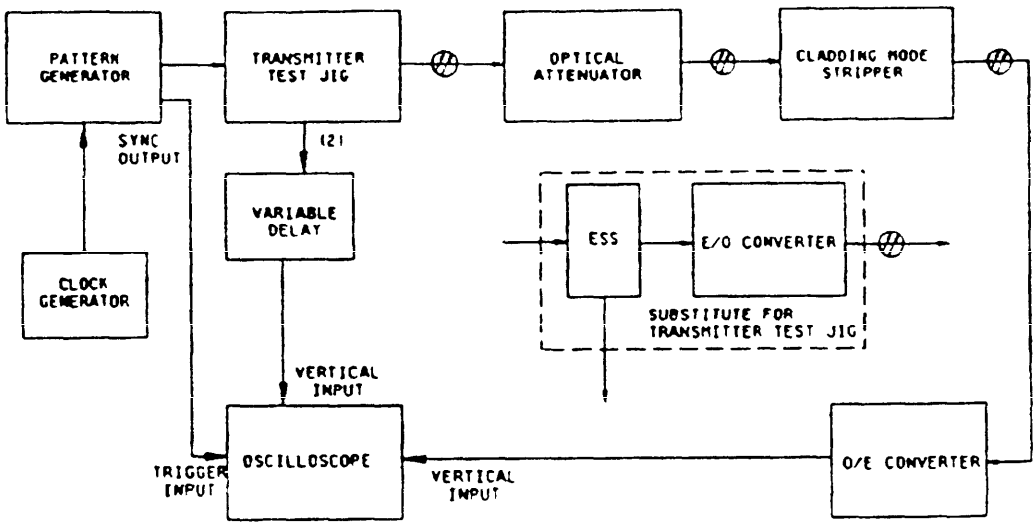
Transmitter delay, D_{tx}, is given by: $D_{tx} = VD_{tx} - VD_{sx} + D_{eo} + D_{ss}$

Receiver delay, D_{rx}, is given by: $D_{rx} = VD_{rx} - VD_{sx} + D_{oe}$

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

Incoming, in-process, and outgoing inventory

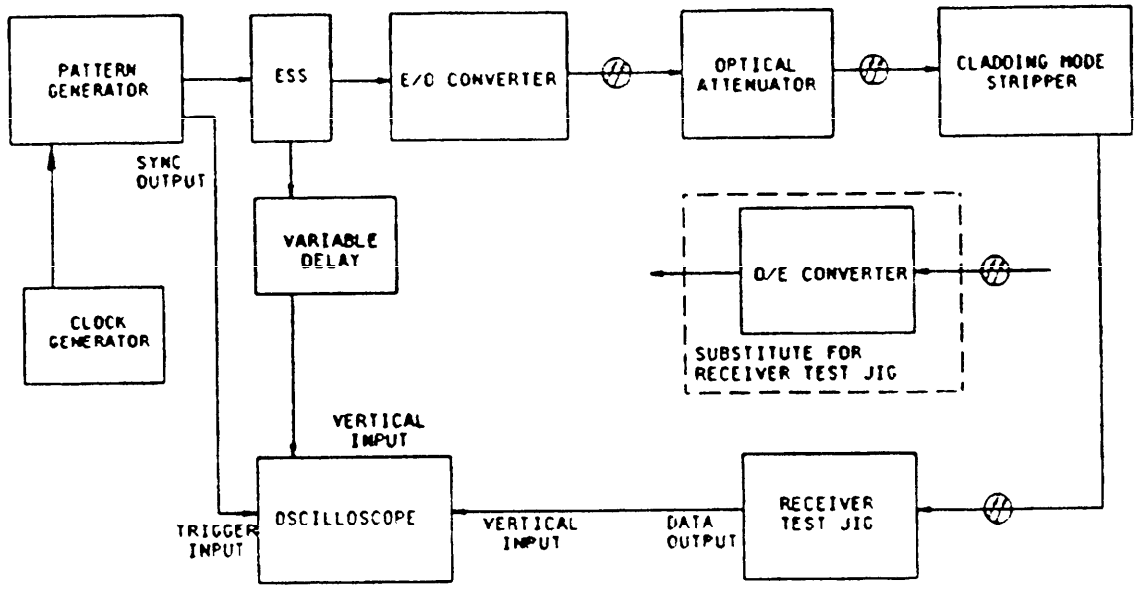
APPENDIX B



NOTES:
1. ESS = electrical signal splitter.
2. Optical carrier modulating waveform sample.

FIGURE 22. Transmitter optoelectronic delay measurement.

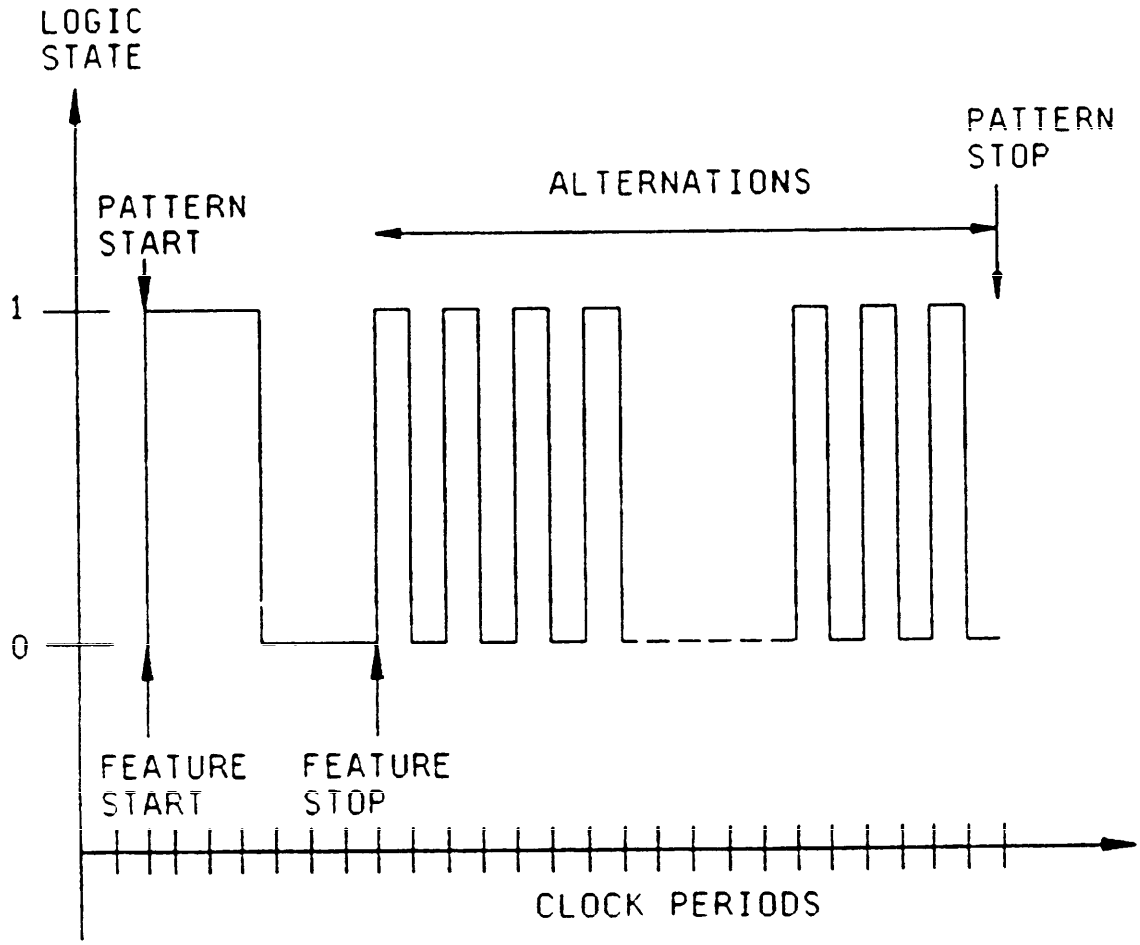
Source: https://assist.dla.mil -- Downloaded: 2026l0-20T02:42Z
Check the source to verify that this is the currentversion before use.



NOTE: ESS = electrical signal splitter.

FIGURE 23. Receiver optoelectronic delay measurement.

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.



NOTES:
1. The portion of the pattern between the feature stop time and the pattern stop time is filled with alternations, a pattern consisting of changes between logic levels "1" and "0" in succeeding clock periods.
2. The length of the pattern, from start time to stop time, is greater than twice the delay to be measured.

FIGURE 24. Optoelectronic delay test pattern.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX B

30.11 FLAG PARAMETER MEASUREMENTS

30.11.1 Scope. This test method describes procedures to be used to evaluate flag outputs from digital fiber optic transmitters and receivers.

30.11.2 Digital receiver signal detect flag output.

30.11.2.1 Introduction. The signal detect flag output indicates whether the digital receiver input power is within a specified range. The flag output is in the "assert" ("deassert") state when the input power is within the requiring specification's "acceptable" ("unacceptable") range. The input power thresholds at which the flag output changes from one state to the other are determined by slowly varying the input power to the receiver under test while observing the status of the signal detect output. Response time is determined by switching input power between levels at which signal detect is required to be in the assert state and deassert state.

30.11.2.2 Symbols used.

A_{ax}: Attenuation setting at the which signal detect flag changes from the deasserted to the asserted state for the modulation rate designated by "x"; l = lowest, m = geometric mean between highest and lowest, h = highest

A_{dx}: Attenuation setting at the which signal detect flag changes from the asserted to the deasserted state for the modulation rate designated by "x"; l = lowest, m = geometric mean between highest and lowest, h = highest

D_{nt}: Measured difference between the arrival times of signals from the O/E converter output and the receiver test jig data output at the vertical inputs to the oscilloscope

D_{pr}: Receiver propagation delay

D_s: Portion of the delay between the signals from the O/E converter output and the DUT data output as seen at the vertical inputs to the oscilloscope which is due to the setup

DUT: Device under test

H(max): Maximum hysteresis measured between assert and deassert levels

H(min): Minimum hysteresis measured between assert and deassert levels

30.11.2.3 Test equipment.

30.11.2.3.1 Required. Test equipment required by this test method is that listed for the "power supply voltage and current" test method (see 30.2), the "basic setup for digital receiver tests" (see 30.4) and in table XIV. The capabilities required of test equipment associated with the referenced test method may differ from those identified herein (see table XIV and 30.11.2.3.2).

Table XIV. Required test equipment.

Item	Generic name	Capabilities	Suggested Mfr/Model
01	Clock generator		
02	Pattern generator		
03	Two level E/O Converter		
04	Electrical signal splitter		
05	Square wave generator	Toggle mode;	
06	Optical fiber coupler 1	Two equal outputs	
07	Optical fiber coupler 2	10% and 90% outputs	
08	Optical power meter		
09	O/E converter	Analog; dc response	
10	Optical attenuator	Resolution of 0.05 dB;	
11	Cladding mode stripper		
12	Receiver test jig	see 30.4	
13	Oscilloscope		Tektronix model OCP5502

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

30.11.2.3.2 Two level E/O converter. The output from the two level E/O converter is an optical carrier binary intensity modulated by the signal at the modulation input and switched between two peak power levels in response to the signal at the switch input. A calibrated adjustment sets the ratio of the two power levels. This setting shall provide resolution of 0.05 dB or better. The discussion herein assumes that the set level difference adjustment affects the lower output level only. The time required for the two level E/O converter to switch between power levels shall be no greater than one tenth of the maximum allowed for the DUT signal detect response. The two level E/O converter may be realized in a number of ways. For example:

a. Use two identical E/O converters with binary intensity modulation and output gating (ON/OFF) capability. Combine their outputs with an optical fiber coupler. Apply the same signal to the intensity modulation input of both sources and adjust phasing so that the modulation envelope waveforms are in phase at the coupler output. Gate one of the sources to produce the required signal.

b. Use an E/O converter with binary intensity modulation capability that employs a current mode switch and current source to drive the optoelectronic transducer. Switch the magnitude of the source current between two values to produce the required signal. Effects associated with extinction ratio and prebias current need to be considered.

c. Use an E/O converter with analog intensity modulation capability and a modulation frequency response extending to 0 Hz. Switch the peak to peak amplitude and average level of the electrical signal at the modulation input to produce the required optical output signal. Effects associated with extinction ratio and prebias current need to be considered.

30.11.2.4 Setup considerations. The considerations and procedures of 30.1 and 30.2 shall be observed or executed, as appropriate. The setup shown in figure 25 shall be used. Unless otherwise specified herein or by the requiring document set the clock frequency and pattern generator pattern to produce a square wave at the highest modulation rate specified for the DUT at the pattern generator output and set the square wave generator to 0 Hz (toggle mode). These procedures require information obtained by performing procedures contained in the "digital receiver BER measurement" (30.8) and "optoelectronic delay" (30.10) test methods.

30.11.2.4.1 Power measurements. The optical power measurement procedure described in the "basic setup for digital receiver tests" (see 30.4) test method shall be used to correlate power meter indications with receiver input power levels.

30.11.2.4.2 Setup delay. The difference between the times required for the two level E/O converter output signal to propagate to the inputs to the receiver test jig and the O/E converter plus the additional delays associated with the outputs from those devices and display by the oscilloscope, Ds, should be known with an error less than 1% of the maximum signal detect response time allowed by the requiring document. This procedure determines that delay (with a slight error). Set the pattern generator to produce a unique and easily identified feature occurring at intervals three times that of the expected delay difference. Toggle the square wave generator output to produce the greater output level from the two level converter. Set the optical attenuator so that the level at the input to the receiver test jig is equal to that at which receiver sensitivity is specified. Connect the O/E converter output and receiver test jig data output to the oscilloscope. Display the waveforms associated with these signals and measure the extent of the time interval between the occurrence of the unique feature in each. Note its value, D_{sm}. The mesial level for the waveform of the O/E converter output signal shall be at the 50% level of that waveform and the mesial level for the data output signal shall be at the reference level for the associated logic family. Subtract the receiver propagation delay, D_{pr}, (see 30.10) from the measured amount. Then the portion due to the setup, of the delay, as measured at the oscilloscope, between these two signals, D_s, is:

$$D_s = D_{ms} - D_{pr}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

30.11.2.5 Procedures.

30.11.2.5.1 Assert/deassert levels. Toggle the square wave generator to produce the greater output level from the two level E/O converter. Monitor signal detect status with the oscilloscope. Increase optical attenuation until signal detect becomes deasserted and then decrease attenuation until it again becomes asserted. Note signal detect state with respect to input power and compare to the phasing specified by the requiring document. Increase attenuation slowly (rate less than 1 dB /sec) to the point at which signal detect changes from asserted to deasserted. Note the setting, A_{dh}. Slowly decrease attenuation to the point at which signal detect changes from deasserted to asserted. Note the setting, A_{ah}. Repeat this procedure with clock frequency set to produce the lowest modulation rate, (A_{dl} and A_{ah}), and again with the clock frequency set at the geometric mean of the highest and lowest modulation rates (A_{dm} and A_{am}). Determine the greatest and least difference between associated assert and deassert levels, that is

H(max) = the greater of (A_{ah} - A_{dh}) or (A_{am} - A_{dm}) or (A_{al} - A_{dl})

H(min) = the lesser of (A_{ah} - A_{dh}) or (A_{am} - A_{dm}) or (A_{al} - A_{dl})

Determine the optical power levels at the input to the DUT associated with the measured values of A_{ax} and A_{dx}. Compare input powers and H(min) with limit values specified by the requiring document.

30.11.2.5.2 Response time measurement. Toggle the square wave generator to produce the greater output from the two level E/O converter. Adjust the optical attenuator to produce the minimum input level to the DUT at which signal detect is required to be asserted. Toggle the square wave generator to produce the lower output from the two level E/O converter. Adjust the two level E/O converter level difference adjustment to produce the maximum input level to the DUT at which signal detect is required to be deasserted. Set the square wave generator period to four times the maximum allowed response time for the DUT. Connect the O/E converter output and the receiver test jig signal detect output to vertical inputs to the oscilloscope. Display the waveforms of these signals on the oscilloscope. Measure the time between a transition of the envelope waveform of the O/E converter output signal between power levels and the corresponding signal detect transition for each envelope transition polarity. The mesial level for the envelope waveform shall be at the 50% level between the levels corresponding to the two power levels and the mesial level for the signal detect output signal shall be the reference level for the associated logic family. Repeat the measurement with the clock frequency set to produce the lowest modulation rate specified for the DUT and for a modulation rate equal to the geometric mean of the lowest and highest rates. The response time is the measured delay less the delay difference determined in 30.11.2.4.2 above. Compare the measured response times with the limit value specified by the requiring document.

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T02:42Z
Check the source to verify that this is the currentversion before use.

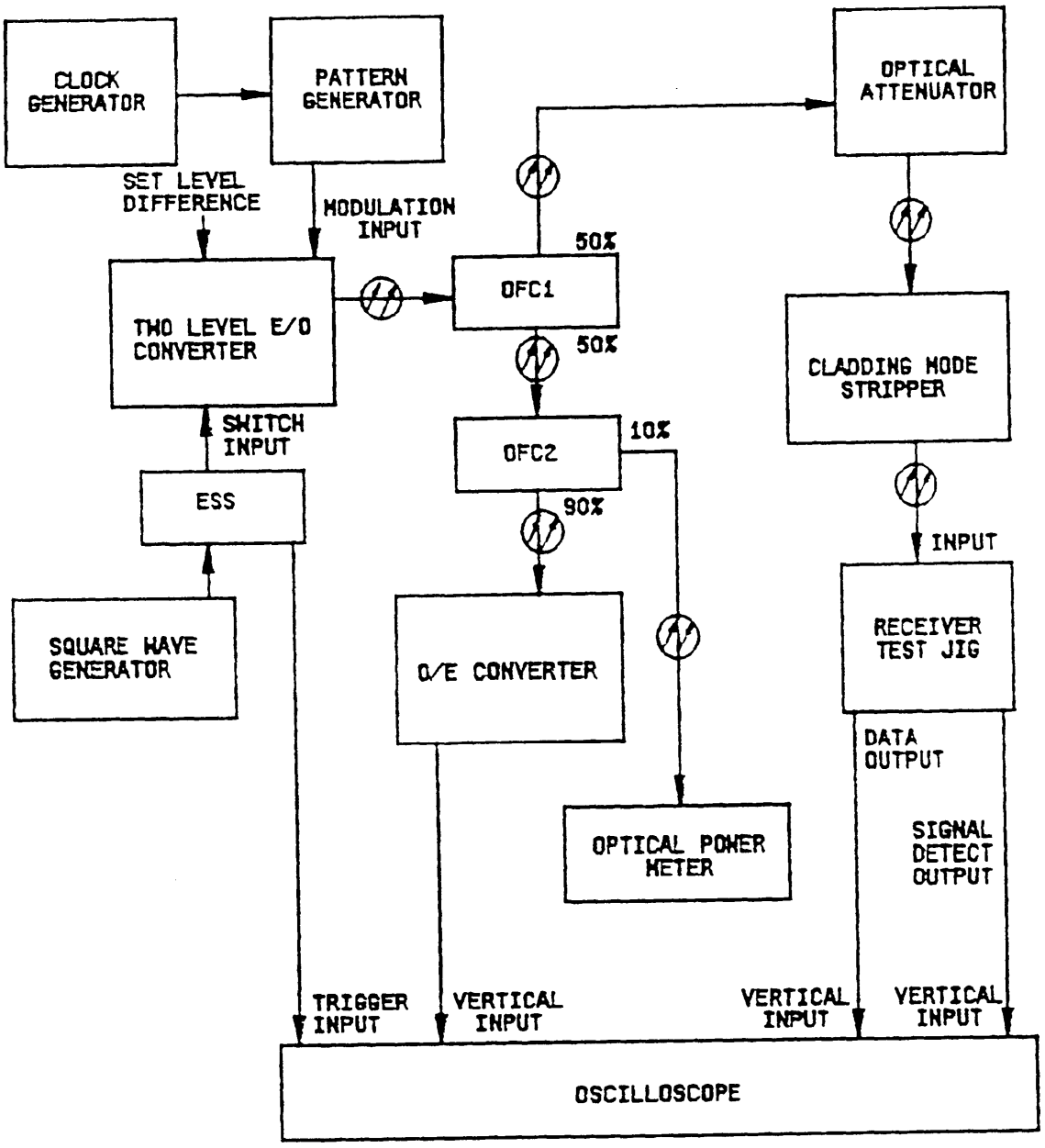


FIGURE 25. Signal detect measurements setup.

Source: https://assist.dla.mil -- Downloaded: 20260-20T02:42Z
Check the source to verify that this is the current version before use.

30.12 **CONTROL PARAMETER MEASUREMENTS**

30.12.1 Scope. This test method contains procedures to be used to evaluate control functions of fiber optic transmitters and receivers.

30.12.2 Transmit disable function.

30.12.2.1 Introduction. This function provides for controlling digital transmitter optical output, enabling or disabling that output independent of DATA input signals.

30.12.2.2 Symbols used.

DUT: Device under test

30.12.2.3 Test equipment.

30.12.2.3.1 Required. Test equipment required by this test method is that listed for the "power supply voltage and current" test method (see 30.2), the "basic setup for digital transmitter tests" test method (see 30.3), and in table XV. The capabilities required of test equipment associated with the referenced test methods may differ from those identified therein (see table XV and 30.12.2.3.2).

Table XV. Required test equipment.

Item	Generic name	Capabilities	Suggested Mfr/Model
01	Oscilloscope		
02	Pulse generator	Toggle mode	

30.12.2.3.2 Considerations. The transmitter test jig should supply a sample output for the disable input signal similar to that for the data input signal.

30.12.2.4 Setup. The considerations and procedures of 30.1, 30.2 and 30.3 shall be observed or executed, as appropriate. The basic setup for digital transmitter tests of 30.3 is used with an oscilloscope and pulse generator connected as shown in figure 26. Set the clock generator frequency and pattern generator pattern to produce a square wave at the pattern generator output at the maximum modulation rate specified by the requiring document for the DUT. Set the pulse generator output top and base voltage levels as appropriate for the transmitter test jig transmit disable input. Set the oscilloscope for "auto trigger" and select the disable input sample output from the transmitter test jig as the trigger source. Display the waveforms due to that signal and the O/E converter output.

30.12.2.5 Procedures. Set the pulse generator to 0 Hz and toggle the output to disable and then enable the DUT. Note the O/E converter peak output level displayed by the oscilloscope for each state. Adjust the oscilloscope to establish these enable and disable peak output levels at convenient vertical levels on the display. The difference between these levels represents the peak to peak amplitude of the envelope waveform. Set the pulse generator width and repetition rate adjustments so that the levels associated with disable and enable states are maintained for durations equal to three times the associated maximum allowed response times. Select oscilloscope trigger polarity to display the control transition associated with the response time to be evaluated (that is, enabled to disabled or disabled to enabled). Set the time base so that the horizontal extent of the display is 1 to 2.5 times the maximum allowed for the response time being evaluated. Measure and record the time between the points at which the disable input sample waveform and the waveform of the envelope of the O/E converter output signal cross their associated reference levels. Unless otherwise specified, the reference level for the disable signal is the reference level for the associated logic family and the reference level for the envelope waveform is at 90% of its peak to peak amplitude for the enable response time and 10% of its peak to peak amplitude for the disable response time. Select the opposite oscilloscope trigger polarity and repeat the preceding steps to determine the response time for the opposing transition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

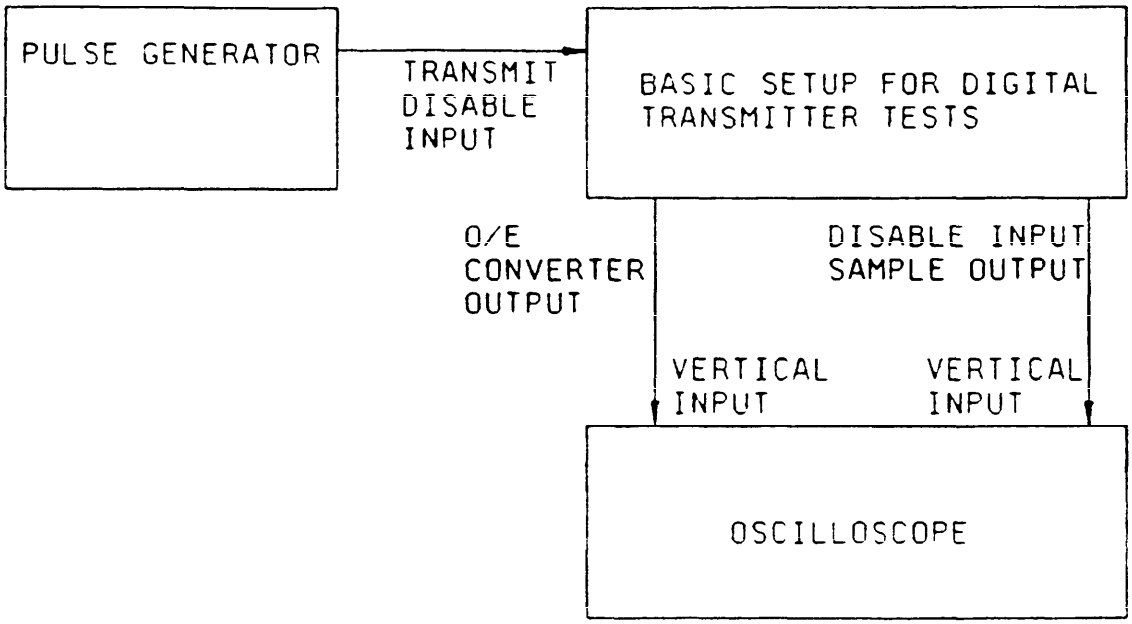


FIGURE 26. Transmit disable measurements setup.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T02:42Z
Check the source to verify that this is the current version before use.

APPENDIX C

PACKAGE OPTIONS

10. SCOPE

10.1 Scope. This appendix describes the requirements for material, construction, and dimensions of the various package styles and optical connections for digital fiber optic devices. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS.

20.1 Specifications. The following specifications form a part of this appendix to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. See 6.2.

SPECIFICATIONS

MILITARY

MIL-F-49291/6 - Fiber, Optical, 62.5/125 Micrometers, Radiation Hardened (Metric)

30. STYLE DESCRIPTIONS

30.1 Style designator. The style designator is provided by the referencing document and consists of two alphanumeric characters as identified and described by the subparagraphs to this paragraph.

30.1.1 Case. There is one type of case. The designator is listed below.

A - Dual-in-line (see 30.2)

30.1.2 Optical connection. There are three types of optical connection. The designators are listed below:

A - multimode pigtail

30.2 Style AA, dual-in-line case with multimode pigtail.

30.2.1 Case A. Case A features are described in 30.2.1.1 through 30.2.1.3. Other requirements are specified in 3.4.2.4 and if applicable, the associated detail specification.

30.2.1.1 Material. See 3.4.1.

30.2.1.2 Outline/dimensions. Case dimensions, exclusive of the optical connection shall not exceed 12.7 mm (0.5 inches) in height, 1.5 inch in width and 2.5 inches length.

30.2.1.3 External finish. See 3.4.2.4.2.

30.2.2 Electrical terminals. Electrical terminal features are described in 30.2.2.1 through 30.2.2.5. Other requirements are specified in 3.4.2.5 and if applicable, the associated detail specification.

30.2.2.1 Type and length. Electrical terminal type and length shall be suitable for mounting on 0.063 inch thick circuit boards.

30.2.2.2 Position. Electrical terminals shall be available through a side defined by the length/width dimensions.

30.2.2.3 Arrangement. Electrical terminals shall be arranged in two parallel rows. In-row terminal spacing shall 0.1 inch on center. Row separation width is left to the contractor.

30.2.2.4 Solderability. See 3.4.2.5.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.

30.2.2.5 Finish. See the associated detail specification.

30.2.3 Optical connection A. Optical connection A features are described in 30.2.3.1 through 30.2.3.3. Other requirements may be specified in the associated detail specification.

30.2.3.1. Type. A pigtail cable shall be as specified in 3.4.2.6.1.

30.2.3.2 Fiber. Multimode fiber within the pigtail cable shall be in accordance with the applicable requirements of MIL-F-49291/6 and as specified in 3.4.2.6.1.1.

30.2.3.3 Position. The optical connection shall be attached to the case on the side defined by the width/height dimensions.

CONCLUDING MATERIAL

Custodians:
Navy - SH
Air Force - 85
Army - CR
NASA - NA

Review activity:
Navy - EC, OS, YD
DLA - ES

Preparing activity:
Navy - SH

Agent:
DLA - ES

(Project 6030-N006)

Source: https://assist.dla.mil -- Downloaded: 202610-20T02:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-24791	2. DOCUMENT DATE (YYMMDD) 95/05/25

3. DOCUMENT TITLE

MODULE, FIBER OPTIC, TRANSMITTER OR RECEIVER, DIGITAL, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, first, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Kendall A. Cottongim	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-5373 (2) DSN 986-5373
ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-ELDT Dayton, Ohio 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Naval Sea Systems Command SEA 05Q42, Washington, DC 20362-5160 Telephone (703) 602-9137/6020 DSN 332-9137/6020

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:42Z
Check the source to verify that this is the current version before use.